Q-COMEM: QUANTIZED SPLIT REPLAY FOR MEMORY-CONSTRAINED HYBRID LLMS

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ABSTRACT

Repeated queries over stable documents create a capacity problem: exact prefix caching minimizes warm latency but retains state at every layer, whereas dense recomputation retains nothing and reprocesses the document. Q-COMEM extends CoMem split replay to hybrid LLMs: an offline Write group-quantizes the depth-j boundary residual with the lower-layer KV, convolution and recurrent state later queries need, and online Read dequantizes and forks it, keeps mutable request state private, and replays the query before reconstructing the suffix, under a fail-closed FORKAUDIT ownership trace, because output equality alone can miss state aliasing. We report *total* memory rather than retained state alone, $M_{\text{total}}(D, N) = S\,D + T(N)$ for D warm documents and N concurrent requests, both terms measured on Qwen3.5-35B-A3B and H20-3e. On an archival 60-item Qasper/2WikiMQA cohort a frozen depth-7 Q4/Q4/Q8 policy cuts the retained term S from 106.235 to 9.661 MiB/document against a like-for-like all-BF16 reference ($10.9965\times$); against the 136.235 MiB the stack retains in native dtypes — 30.000 of it FP32 GDN recurrent state in excess of BF16 — the ratio is $14.10\times$, so part is dtype narrowing, not packing, and either way it is one term of two. Its mean F1 (0–100) is 54.24 versus 54.68, paired -0.45 $[-2.06, 0.99]$ (10,000 item-level resamples, seed 20260902). The transient term runs the other way: at fanouts $N \in \{1, 2, 4, 8\}$ Q-COMEM's measured peak transient exceeds the exact cache's at every one, 930.91 against 286.44 MiB at $N = 1$ and 1713.15 against 1441.93 at $N = 8$, the fitted lines crossing only at $N = 12.5$, outside the measured range. Their sum names a regime rather than a ratio: Q-COMEM holds the smaller total above roughly five warm documents at low concurrency and two at $N = 8$, by $2.42\times$ at twenty warm documents and $N = 4$, $7.37\times$ at a hundred and $N = 1$, and the larger total below, so a deployment keeping a handful of documents warm while serving many concurrent requests against them is outside its useful range. The model assumes S is linear in D and T independent of D — no run held more than one document resident — and its fits license no N above 8. A composition run shares one packed depth-7 entry across those fanouts under the ownership discipline itself: shared mode effective with no fallback, sharing non-vacuous to the end of the request, the $N > 1$ shared traces token-identical to the published $N = 1$ private-materialization path, and every applicable FORKAUDIT target, the three previously untested packed-entry obligations included, covered and passing there. What is shared is small — 2 of 16 tensors and 0.500 of 12.938 MiB per request, about 4% — so it demonstrates the ownership discipline on the packed path, not a material reduction of per-request memory; and the primary FORKAUDIT factorial still runs full-prefix BF16 with no split or quantization, so ownership on the exact code path behind the deployment tables transfers by argument. The costs are explicit. Against the only quantized exact cache we measured the frozen policy is $5.84\times$ smaller at 3.41 points higher mean F1 (paired -3.4122 $[-8.4296, -0.0390]$, excluding zero by 0.0390 — enough for the ordering, not to call the effect large or robust), but that ratio is width-confounded, factoring into $3.918\times$ split and $1.491\times$ bit width, and no width-matched exact cache was executed; this is a selected trade-off point, not selection-corrected, and not a statistical-equivalence claim. Timing is adverse throughout: exact reuse, honest dense recomputation and two published same-stack prefix caches all reach the first token sooner, throughput roughly halves, and the per-repeat spread is wide enough — generation stopping after four to

five tokens against a 32-token cap, one TPOT maximum of 404.01 ms against a 53.45 ms median, a throughput column reconciling with no generated-token count — that we print minimum, median and maximum per timing cell and read this evidence as weaker, not stronger. The supported contribution is capacity reduction with an explicit online cost, not TTFT acceleration. The 54.5%/42.2% allocator saving of the ownership component is measured against a full-copy control, a controlled ownership factor and not an optimized baseline: against this paper's own paged-prefix baseline the audited fork ties and is worse on generation increment, buying auditable request-local state, not fewer allocator bytes. The audit exposes a historical alias that output and terminal-state equality miss but a persistent-base content invariant also catches, so its increment is transition-time localization, not exclusive detection. We claim no lower total process memory, measured recall, edge-device performance or broad quality preservation.

1 INTRODUCTION

Repeated queries over stable documents are common in document QA, agents, tree search, and best-of-N decoding. Reusing a prefix avoids repeated prefill, but exact caching retains per-document state at every layer, competing with weights, request workspace, and other resident documents. The question is capacity-first: how much reusable state must stay resident, and at what cost? Dense recomputation and exact caching are the endpoints; CoMem instead writes a document to an intermediate depth and reuses the per-token residual (Liu et al., 2026a;b). Hybrid LLMs make this harder: lower full-attention layers also need KV, lower linear layers convolution and recurrent state, and those mutable states cannot be shared blindly across requests (Yang et al., 2025; Lenz et al., 2025).

Q-COMEM is a quantized hybrid split-replay design (Figure 1): offline Write packs the split residual with the lower-layer state a future query needs under state-type-aware low-bit widths; online Read dequantizes it, shares only immutable document tensors, creates request-local mutable state, and reconstructs the suffix, spending computation to save retained bytes. Because equal outputs can hide a mutable-state alias, we pair it with FORKAUDIT, a phase-aware ownership trace over immutability, copy-on-write (COW), recurrent rebinding and call provenance, evaluated on a full-prefix BF16 factorial and separately on the packed Read path. We contribute:

- **Whole-entry accounting for a hybrid split-replay entry.** Group-wise asymmetric quantization and per-layer, per-state-type width assignment — neither claimed as new (Section 2) — applied to an entry those methods do not address: the depth-j residual *together with* the lower full-attention KV and convolution/recurrent state a hybrid backbone needs. CoMem retains only the residual, unquantized in BF16, so every number here uses the whole-entry denominator (Section 4.1).
- **Total memory, not the retained term alone.** We measure both terms of Eq. 1 a method controls — retained state per warm document, peak transient allocation per query — and report their sum $M_{\text{total}}(D, N) = S\,D + T(N)$: the frozen Q4/Q4/Q8 point is $10.9965\times$ smaller on S ($14.10\times$ against the native-dtype entry) at -0.45 $[-2.06, 0.99]$ mean F1, *larger* on T at every fanout measured, and ahead on the sum only above roughly five warm documents at low concurrency and two at $N = 8$ (Sections 5.2, 5.6). A quantized exact cache is $5.84\times$ larger and 3.41 points lower on a ratio that is not width-matched, and timing is adverse against in-house and published same-stack systems alike, under per-repeat dispersion we print rather than average away (Sections 5.3–5.4). Retained payload stays separate from allocator and process memory, and Recall was measured in no cohort.
- **Ownership-safe reuse, now audited on the packed path too.** The seven-target FORKAUDIT contract couples semantic relations to phase-indexed storage evidence across 96 ownership configurations, all full-prefix BF16 KV with no split depth and no quantization; a separate composition run shares one packed $j = 7$ Q4/Q4/Q8 entry across $N \leq 8$ concurrent requests, covering all ten targets — the seven plus the three packed-entry obligations — at token identity with the published private-materialization path while sharing 4% of per-request state, so it demonstrates the discipline and not a memory saving (Section 5.6). A historical alias corrupts the persistent base while output and terminal request state stay exact, but a persistent-base content invariant also catches it, so the increment is transition-time localization, not exclusive detection. Against our own paged-prefix baseline the audited fork

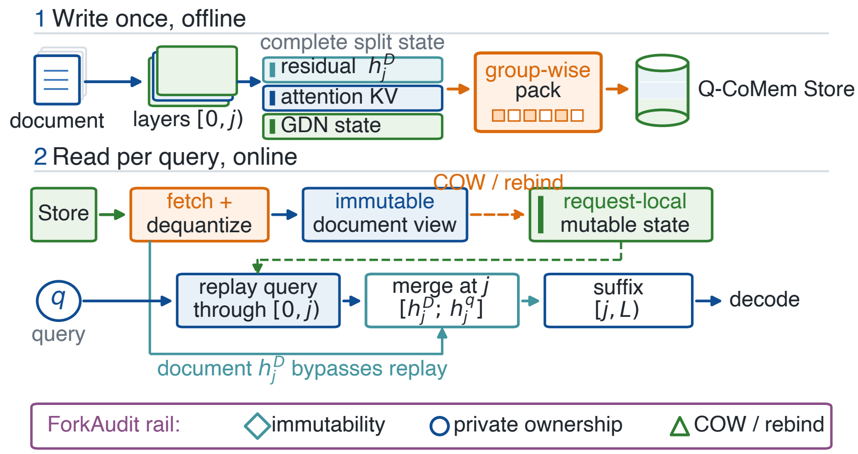

Figure 1: Q-COMEM protocol geometry. Write packs the depth-j residual with the lower attention KV and GDN convolution/recurrent state; Read dequantizes it, exposes immutable document tensors, creates request-local mutable state through COW/rebinding, and replays the query through $[0, j)$ before the suffix. The FORKAUDIT rail at the foot validates the ownership boundary and is not an execution stage; the primary factorial ran full-prefix BF16 with no split depth and no quantization, so the packed Read path drawn here is covered only by the separate eight-item composition run of Section 5.6, and not on the code path behind Tables 4 and 22.

ties on both allocator endpoints and is worse on generation increment; the 54.5%/42.2% reduction at $N = 32$ is against a full-copy control, not that baseline (Section 5.5).

The timing execution uses an eight-item subset under its own generation budget, the operating-point run a separate execution, the ownership factorial a third system entirely and the composition run a fourth (Sections 5.1, 5.6); "Store" is the deduplicated sum of the tensor-storage bytes one entry owns, not total VRAM, process memory, active workspace, or admitted capacity (Sections 5.1, 5.7).

2 RELATED WORK

CoMem persists one depth-j residual per document token and replays the upper suffix for later queries (Liu et al., 2026a;b); we build on that interface rather than claiming a new split-replay primitive, and add quantizing the residual *and* the lower KV/recurrent state a hybrid model needs under an explicit storage denominator. PagedAttention, Prompt Cache, RadixAttention, ChunkAttention, Hydragen and Preble supply the block sharing and prefix reuse we take as the low-TTFT reference (Kwon et al., 2023; Gim et al., 2024; Zheng et al., 2024; Ye et al., 2024; Juravsky et al., 2024; Srivatsa et al., 2025); Gated Delta Networks, Jamba, Marconi and HYPIC supply the hybrid architectures and reusable hybrid prefixes, stored unquantized (Yang et al., 2025; Lenz et al., 2025; Pan et al., 2025; Liu et al., 2026c). Low-bit storage of reusable state is likewise established prior art and Q-COMEM claims none of its mechanisms: the packer of Section 4.2 is the standard asymmetric affine quantizer (Jacob et al., 2018) applied group-wise, the form FlexGen used (Sheng et al., 2023), and $\mathbf{b}$ is one setting on an established grouping and mixed-precision surface, not a new selection method. XQuant is the closest prior work to our core mechanism: it caches quantized layer-input activations and rematerializes keys and values during decode (Tomar et al., 2025), so persisting a quantized intermediate and paying recomputation to avoid retaining downstream state is already its thesis. What differs is the object — a depth-j boundary residual rather than a layer input, and a heterogeneous hybrid entry rather than one state family, raising the ownership question of Section 4.3. We claim the hybrid entry and its accounting, not quantize-and-recompute. Appendix G attributes each surface work by work, states both XQuant differences in full, and gives Tables 20 and 15.

3 MOTIVATION

A deployment must fit model weights, shared adapters, active workspace, retained document entries, and runtime state at once:

$$M_{\text{device}} \geq M_{\text{weights}} + M_{\text{shared}} + M_{\text{active}} + \sum_{D \in \mathcal{H}} S(D) + M_{\text{runtime}}, \tag{1}$$

where $\mathcal{H}$ is the hot document set. Weights and runtime are fixed once a model and stack are chosen, and Eq. 1 multiplies the retained term by $|\mathcal{H}|$: the endpoints are dense recomputation, $S(D) = 0$ at full prefill per query, and exact caching, 136.235 MiB/document here, with split replay at depth j between them at the cost Section 5.3 measures. The workspace term is not method-independent, and our own Read path is why: it materializes a private dequantized copy of the entry per concurrent request (Section 4.3), so M_{active} scales with the entry a method chose to retain. We therefore do not report the retained term by itself: Section 5.6 measures M_{active} for both arms and reports the sum, what a deployment budgets. Q-CoMem still targets $S(D)$ alone: it does not compress weights or promise a smaller workspace, reports allocator measurements under a separate denominator, and leaves process/NVML memory unmeasured.

In a hybrid model that entry is not just larger but mutable: an honest denominator counts the whole entry, and because those states are mutated in place by the kernels that consume them, sharing one entry across concurrent requests is an ownership question output comparison cannot answer (Section 5.5).

4 METHODOLOGY

4.1 RETAINED STATE DECOMPOSITION

For split depth j the reusable hybrid state is the boundary residual h_j^D, lower-layer full-attention KV, and lower-layer convolution/recurrent state; with quantizer Q_b the retained payload of Eq. 1 is

$$S_{\text{QCoMem}}(D; j, \mathbf{b}) = |Q_{b_h}(h_j^D)| + \sum_{\ell < j, \ell \in \mathcal{A}} |Q_{b_\ell}(K_\ell^D, V_\ell^D)| + \sum_{\ell < j, \ell \in \mathcal{G}} |Q_{b_\ell}(C_\ell^D, G_\ell^D)|, \tag{2}$$

where $\mathcal{A}$ and $\mathcal{G}$ index full-attention and recurrent layers, and $\mathbf{b}$ is assigned per state type and per layer. Counting only h_j^D would understate the deployed state.

4.2 OFFLINE WRITE: PACKING THE COMPLETE ENTRY

The document runs through layers $[0, j)$; we retain h_j^D and the lower cache leaves a later query needs, partitioning each floating tensor into 64-value groups. For a b-bit group with minimum m and maximum u we store

$$s = (u - m)/(2^b - 1), \qquad q = \text{clip}\big(\text{round}((x - m)/s), 0, 2^b - 1\big), \tag{3}$$

with packed unsigned values and BF16 scale/bias metadata. The equation omits three reimplementation details — the grouping axes and their padding, the degenerate constant group, the FP32 reconstruction path — given in Appendix C.

We write *native dtype*, not “Q16”, for the unpacked reference, because that arm asserts no uniform width: BF16 residual and attention KV, the stack’s FP32 GDN recurrent state. Every policy is one frozen bit vector named here, each a Q4 residual with cache widths: *frozen Q4/Q4/Q8* $[8, 8, 8, 4, 8, 8, 8]$, *calibrated per-layer* $[8, 8, 4, 4, 8, 8, 8]$, and *aggressive mixed* $[8, 8, 2, 2, 2, 8, 2]$, the last introducing the two-bit width. Their calibration set, byte budgets, and separation from the unchanged BF16 model layers and forward computation (Figure 2) are in Appendix C. Each is one frozen operating point, not a claim that layer-wise selection is optimal.

4.3 ONLINE READ: OWNERSHIP-SAFE FORK AND SUFFIX REPLAY

For each query the packed entry is dequantized into a logical document view, whose logical fork for resident requests $r \in \{1, \ldots, N\}$ is $\mathcal{F}_r(D) = \big(\{K_\ell^D, K_\ell^r\}_{\ell \in \mathcal{A}}, \{G_\ell^D, G_\ell^r\}_{\ell \in \mathcal{G}}\big)$, where document state K_ℓ^D, G_ℓ^D is immutable and may be shared, whereas append reservations K_ℓ^r and mutable recurrent state G_ℓ^r are request-local. A borrowed recurrent base is read-only at setup and must rebind to private storage when the registered transition first mutates it, and a partial KV tail is copied before append. This discipline is what FORKAUDIT audits; the Transformers implementation behind both deployment tables instead materializes a full private copy of the dequantized entry per query, sharing nothing and exercising neither borrowing nor copy-on-write. Section 5.6 implements and audits the borrowing path itself at $N \leq 8$, token-identically to that private path. Query tokens then traverse the lower layers using request-local state; document and query residual chunks — distinct, as the Qwen3.5 recurrence requires — seed the suffix through $[j, L)$ before greedy decoding.

4.4 FAIL-CLOSED VALIDATION OF THE OWNERSHIP CONTRACT

FORKAUDIT validates the ownership discipline of Section 4.3 with seven targets — frozen identity, prefix immutability, private ownership, tail-safe COW, bounded host-side call provenance, cross-arm semantic equivalence, cross-N prefix consistency — and three more specific to a packed entry: dequantized-view immutability, residual-chunk binding, packed-entry lifetime. The primary factorial runs full-prefix BF16 with no split depth and no quantization (Section 5.5) and covers the first seven; the composition run of Section 5.6 covers all ten on the dequantize-then-fork Read path, at one entry, one depth, one bit vector and $N \leq 8$. A target passes only if its mandatory-slot coverage is complete, every slot binds, and every replay predicate holds, so coverage is separate from verdict (Appendix E). Targets 2–5 are pre-output storage/call obligations; 6–7 compare registered semantic observables and cannot substitute for ownership evidence. It is an offline, non-adversarial regression contract, not security attestation: capture/replay, the mandatory-slot schema, slot-to-live-tensor binding and framework storage semantics form the trusted computing base (TCB), and a pass establishes only that the registered predicates held at captured points under it, covering neither coherent omission/fabrication and transient writes between observations nor compiled GDN, ATen/CUDA, or device/driver execution.

5 EXPERIMENTS

5.1 SETUP AND MEMORY DENOMINATORS

Quality cohort and generation budgets. The broadest Store–F1 panel is an archival, raw-first validation run over 60 paired items — Qasper and 2WikiMQA source indices 6–35, 30 per dataset — under official LongBench-v1 prompts and a 4,096-token input cap (Bai et al., 2024a; Dasigi et al., 2021; Ho et al., 2020). Decoding is greedy with an EOS stop and a per-dataset budget of 128 new tokens for Qasper and 32 for 2WikiMQA; the operating-point run uses the same budget and the eight-item panel 32 for both, so the three panels' F1 levels are not interchangeable. Four calibration prompts (indices 4–5) precede and are disjoint from validation; the shards record no timing or Recall, and the interpreter version was not recorded. Every execution here runs Qwen3.5-35B-A3B on H20-3e under the toolchain Table 22 names; a separately executed *operating-point run* covers the same 60 items with one warm-up, three repeats and seed 20260903, recording TTFT and TPOT and adding a quantized exact cache (full-prefix Q8: the same codec on the unsplit entry at a uniform 8-bit width, *not* the frozen policy's bit vector) and a $j = 13$ split control.

F1 is our reimplementation of the LongBench-v1 `qa_f1_score` protocol, not the upstream module, and two behaviours deviate — an empty prediction or reference scores $\mathrm{float}(\text{pred tokens} = \text{ref tokens})$, and punctuation stripping is ASCII-only — which we state because every F1 value, delta and interval here depends on it; (Bai et al., 2024b) is the protocol source, not the code that produced these numbers.

Latency and composition executions. A separate registered systems execution uses eight items (Qasper and 2WikiMQA source indices 6–9): a subset of the quality cohort's *items* but not of its measurement protocol, since it caps every arm at 32 generated tokens. That is why its full-prefix F1 reads 39.14 against those panels' 54.68 and 54.67 on a superset of the same items; the levels are not comparable, and we do not decompose the gap into budget and item-difficulty shares, per-item F1 not having been extracted under both protocols. A fourth execution, on those same eight items, shares one packed $j = 7$ entry across $N \leq 8$ concurrent requests (Section 5.6); it is pooled with no other panel and its F1 is not a new quality result.

Memory denominators. In both deployment tables "Store" is the deduplicated sum of the bytes of the tensor storages one retained entry owns (Appendix C), a 60-item mean in Table 4 and an eight-document median in Table 22. It is *not* the entry's physical byte-range union, which the same code computes but does not print; that distinction was previously stated the other way round, and the correction moves no number. The 136.235-versus-140.34 and 9.661-versus-10.01 difference is cohort and statistic, not length — we withdraw the earlier 3.8%-longer-documents explanation — and decomposes into 69.3% cohort, 30.7% mean-versus-median and 0.0% estimand (Appendix C). No Store denominator includes non-tensor/Python metadata, allocator pools, process/NVML memory, active workspace, or admission capacity; Section 5.5 uses a third denominator and Section 5.6's peak transient allocation a fourth — a CUDA-allocator peak above a pre-fork baseline, not a Store sum and not additive with one — and every quality difference uses the full-prefix arm of its own execution. Both deployment tables are Transformers-only executions of packed entries, whereas Table 19 and every verdict of the primary FORKAUDIT factorial come from vLLM 0.26 plus Transformers, full-prefix BF16 KV in 128-token pages, no split, no quantization, PG-19 — so no verdict of that factorial covers those packed entries, which only the composition run covers.

5.2 MEMORY: RETAINED STATE PER DOCUMENT

Retained state is one of the two method-dependent terms of Eq. 1; Section 5.6 measures the other and reports the sum. Under the all-BF16 convention, which isolates packing from the stack's dtype choices, splitting reduces mean Store from 106.235 to 28.683 MiB/document ($3.7038\times$) and the frozen policy to 9.661, $10.9965\times$ below the reference. The entry the stack retains is larger, 136.235 MiB/document, because it keeps GDN recurrent state in FP32; 30.000 of that is excess over a BF16 encoding, and against it the same two steps read $3.93\times$ and $14.10\times$, so part of that physical saving is dtype narrowing rather than the Eq. 3 packing step. Both columns are exact re-analyses of the same archived rows (Appendix C), and both are the retained term alone. The full panel is Table 4; its other retained means are 34.683 MiB/document (native-dtype split), 9.395 calibrated per-layer ($11.3073\times$ all-BF16, $14.50\times$ native), 7.536 aggressive mixed ($14.0969\times$, $18.08\times$) and 0.000 dense, each block's full-prefix row being its own $1.00\times$ reference.

The operating-point run supplies the arm the archival panel lacks: full-prefix Q8, the unsplit entry at a uniform 8-bit width, retains 56.438 MiB/document, $1.8823\times$ below the all-BF16 reference and $2.41\times$ below the native-dtype one, but $5.84\times$ above the frozen point; its $j = 13$ split control retains 16.101 ($6.5980\times$, $8.46\times$). That ratio is not width-matched — the frozen policy holds the residual and its one lower attention layer at 4 bits against the exact cache's uniform 8 — so we report its factors, not the product alone: Appendix C splits it exactly into $3.9174\times$ from split depth and $1.4912\times$ from bit width, and only the former is a split-replay advantage, agreeing with the $3.93\times$ split step above. *No width-matched exact cache was executed*: the 37.85 MiB point is a byte count fixed by Eq. 3's format identity once the bit vector is given, and we make no quality claim for it.

5.3 LATENCY: TTFT, TPOT, AND THROUGHPUT

The eight-item timing characterization is not a speedup over exact caching (Table 22a): median TTFT rises from 0.163 to 0.673–0.674 s on the Q-COMEM rows, which dequantize and rebuild the suffix, and median throughput falls from 13.38 to 7.15–7.19 tokens/s at TPOT within 2.5% of 54.11 ms. On all 60 items (Table 4) Q-COMEM is slower to the first token than honest dense recomputation as well as than exact reuse, while TPOT stays between 54.4 and 56.0 ms at every arm; its recorded TTFT reads 0.636 s dense, 0.181 full-prefix, 0.182 full-prefix Q8, 0.582 at $j = 13$, 0.656 at $j = 7$, and none in the archival run.

What the dispersion does to all of that. Those cells concealed two facts, and readers were right that no generated-token count reconciles the throughput column (Table 22, note †). Panel (b) prints the cause, from the one execution whose per-repeat shards were retained: generation halts after a median of four or five tokens under the protocol's EOS policy, so the throughput denominator is four tokens and not the 32-token cap; and the per-repeat spread is very large, TTFT maxima of 1.09–1.57 s against medians of 0.16–0.67 s, a full-prefix TPOT maximum of 404.01 ms against a 53.45 ms median. Three consequences follow, each against us. The 2.5% TPOT band compares point estimates whose repeat-level dispersion is unmeasured on that panel and, where measured, far outside the band, so it is not a tolerance. No TTFT ordering is resolved by three repeats, every pair of min–max ranges in panel (b) overlapping. And the ordering is unchanged at every aggregation we computed and stays adverse: frozen $j = 7$ is behind both exact caching and honest dense recomputation. We make no latency claim, and record that this dispersion makes the timing evidence weaker, not stronger.

Against deployed systems, not only our two in-house endpoints, the comparison is harder. On the identical eight items and hardware, SGLang 0.5.17 Radix reaches the first token in 0.146 s at 15.83 tokens/s and the published HYPIC codebase's prefix cache in 0.072 s at 19.07 tokens/s, both at mean F1 39.14, against 0.673–0.674 s and 7.15–7.19 tokens/s at 39.01–39.14 here (Tables 23, 24; within block). Those arms retain a median 139.53 and 324.09 MiB/document against 15.89 for Q-CoMem split Q8 — three levels we print rather than ratio, for the reason Appendix H.1 gives. The supported result is a retained-capacity trade-off with an explicit online cost and no latency advantage over any endpoint here.

5.4 Answer quality

The frozen Q4/Q4/Q8 point loses -0.45 $[-2.06, 0.99]$ mean F1 against full-prefix (95% percentile intervals, 10,000 paired item-level resamples, seed 20260902), negative on both datasets; the two steps compound rather than cancel, splitting costing -0.06 and quantizing a further -0.39, and no item crosses the registered catastrophic threshold under either arm (0/60 against dense and against full-prefix). That block's mean F1 is 54.68 full-prefix and 54.62, 54.24, 53.36, 49.19, 54.29 for split, frozen, calibrated, aggressive and dense, differing from its 0.00 reference by -0.06 $[-0.18, 0.00]$, -0.45 $[-2.06, 0.99]$, -1.32 $[-4.33, 0.92]$, -5.50 $[-11.58, -0.20]$ and -0.39 $[-4.66, 2.60]$ (Table 4). The headline policy was selected among six on this panel, so the interval is not selection-corrected and proves neither equivalence nor noninferiority (Appendix B).

In the operating-point run the frozen $j = 7$ point is both $5.84\times$ smaller than the only quantized exact cache we measured and 3.41 points higher in mean F1 (54.297 against 50.885, each with its own interval against that run's 54.668 full-prefix reference); that $5.84\times$ is the width-confounded ratio Section 5.2 decomposes, and no width-matched arm exists to say whether the F1 ordering survives at matched precision. The paired interval between those two arms now exists: over the 60 common item keys, full-prefix Q8 minus frozen $j = 7$ is -3.4122 $[-8.4296, -0.0390]$ (seed 20260903), excluding zero, so the ordering is supported on this cohort rather than merely observed. Its five arms read 54.67 full-prefix, 54.30 at $j = 7$ (-0.37 $[-1.97, 1.08]$), 51.27 at $j = 13$, 50.89 full-prefix Q8 (-3.78 $[-9.12, 0.32]$) and 54.18 dense (-0.49 $[-4.57, 2.55]$). We claim that and no more: the upper endpoint is -0.0390, near enough to zero that a different resample sequence could carry it across, so the interval is evidence neither that the effect is large nor that it is robust. Both neighbouring pairs from that pass cross zero, so the frozen point separates from the quantized exact cache and from neither unquantized reference; all ten pairs, with their values and their agreement with the printed rows, are in Appendix B. The $j = 13$ control loses 3.40 $[-8.67, 0.52]$, nine times that loss, fixing the split depth at $j = 7$ for this policy — one parameter, and we report nothing about depths this cohort did not measure.

The two mixed policies are ablations, not improvements *on the panel that selects them*: the calibrated per-layer policy is named for its budget, not measured parity, and the aggressive policy reaches $18.08\times$ with its interval below zero and five threshold-crossing items (Appendix B). The eight-item panel orders the first pair the other way, the calibrated policy being smaller *and* better there (Table 22a) under a different budget and with no interval; the 60-item cohort is the selector because it is larger, carries paired intervals, and uses the full budget, and we report the inversion rather than the favourable half of it. Recall was recorded in no cohort and we make no recall claim.

5.5 Ownership and correctness validation

Scope, cohort, and procedure. The primary factorial runs full-prefix BF16 KV on a vLLM-plus-Transformers stack with no split depth and no dequantization step, and so audits the ownership discipline of Section 4.3 on that configuration, not the quantized Read path of Tables 4 and 22, which Section 5.6 audits separately. Three executions produce these results (Appendix A; Tables 9–10): all seven targets have complete mandatory-trace coverage and pass at their declared fixed-stack scope, the live-binding rerun's stronger owner-row binding covering only one selected cell (Section 5.7). Every paired mutant is rejected at its frozen gate while token equality and exact logits miss most of them, and in a no-injection defect a persistent-base content invariant catches what output and terminal

Table 1: Total memory, $M_{\text{total}}(D, N) = S\,D + T(N)$, for D warm documents and N concurrent requests: the warm-set crossover and four worked deployment points, from the measured retained means S of Table 4 and the measured transient fits $T(N)$ of Table 5 (Section 5.6).

Warm-set crossover D: 5.13 at $N = 1$, 4.69 at $N = 2$, 3.79 at $N = 4$, 2.01 at $N = 8$

Warm documents D	Requests N	Full-prefix (MiB)	Q-COMEM (MiB)	Ratio
4	8	1978.6	1726.4	1.15×
20	4	3493.3	1441.9	2.42×
100	1	13893.2	1885.5	7.37×
100	8	15057.2	2653.9	5.67×

Every cell carries the three unmeasured assumptions of Section 5.6: S linear in D (no run held more than one document resident); T independent of D (unverified); fits valid only for $N \in [1, 8]$, so no cell licenses a statement about $N > 8$. The transient term is an allocator peak on this stack, is not additive with Store, and fixes no admitted capacity. No workload of the many-warm-document shape these rows describe was executed.

state do not (Appendix H), so the increment is transition-time owner/layer/family localization, not exclusive detection. These and Appendix H's cohorts are fixed-case comparisons against registered observables, not a detection study or an error rate.

Allocator endpoints under a controlled ownership factor. Against the appropriate reference the ownership component saves nothing: at $N = 32$ the audited fork ties this paper's own paged-prefix baseline exactly (2.229 GiB final, 2.843 GiB peak) and is 1.950 against 0.019 GiB worse on generation increment, so it buys auditable request-local state, not fewer allocator bytes. The 54.5%/42.2% reduction is against a different arm, a full-copy control that is a controlled ownership factor and not an optimized baseline; Appendix H prints both endpoints.

5.6 COMPOSITION: ONE PACKED ENTRY SHARED ACROSS CONCURRENT REQUESTS

The composed system runs, the contract holds on it, and the sharing is small. A fourth execution composes the two halves, sharing one packed depth-7 Q4/Q4/Q8 entry across $N \in \{1, 2, 4, 8\}$ concurrent requests. The shared mode takes effect with no fallback; sharing is non-vacuous at the policy's window *final*, persisting to the end of the request, not collapsing at the first append; the $N > 1$ shared traces are token-identical, per request, to the published $N = 1$ private-materialization path; and every applicable target — the seven of Section 4.4 plus the three packed-entry obligations named untested there — has complete mandatory-slot coverage and a passing predicate there. Agreement with the full-prefix arm is recorded as a diagnostic and gates nothing, that comparison crossing the document/query chunk boundary the Qwen3.5 recurrence is sensitive to. *What is shared is small*: at the sharing window 2 of 16 tensors and 0.500 of 12.938 MiB per request are shared, about 4% of per-request state. This demonstrates the ownership discipline of Section 4.3 on the packed path, not that sharing materially reduces per-request memory.

Transient allocation refutes an arithmetic prediction of ours. The same run measures peak transient allocation per query (Table 5): Q-COMEM allocates *more* than the exact cache at every fanout measured, 930.91 against 286.44 MiB at $N = 1$ narrowing to 1713.15 against 1441.93 at $N = 8$, the fitted lines $809.68 + 109.76N$ and $103.45 + 166.28N$ crossing at $N = 12.5$, outside the measured range. Sharing helps without closing it: 214.87 MiB saved at $N = 4$ and 263.83 at $N = 8$ against private materialization, the slope cut from 148.38 to 109.76 MiB per request. We had predicted the opposite from the component inventory — smaller intercept and smaller slope, hence no crossing at any N; the slope is indeed smaller, but the intercept is roughly eight times larger, for the reason Appendix C now records. This strengthens Section 5.7's admission that reducing Store does not by itself prove serving capacity.

Total memory, and the regime it names. The quantity a deployment budgets is $M_{\text{total}}(D, N) = S\,D + T(N)$, S the retained state of Section 5.2 (136.235 MiB full-prefix, 9.661 frozen $j = 7$) and T the measured transient. Summing two measurements is admissible, but three unmeasured assumptions travel with it: S is linear in D, structural since entries are independent, though no run held more than one document resident; T does not depend on D, plausible for per-request reconstruction work but unverified; and the fits hold only for $N \in [1, 8]$, four points, so we state nothing about N above 8. Under them Q-COMEM has the smaller total once the warm set exceeds 5.13 documents at $N = 1$, 4.69 at $N = 2$, 3.79 at $N = 4$ and 2.01 at $N = 8$, and the margin grows with it — $2.42\times$ at twenty warm documents and $N = 4$, $5.67\times$ at a hundred and $N = 8$ (Table 1). The other direction is as plain: below the crossover its total is larger, so a deployment keeping a handful of documents warm while serving many concurrent requests against them is outside this method's useful range. The setting it suits is the opposite, many documents warm and each queried intermittently, which we describe qualitatively only, having run no benchmark, task or trace of that shape. The $14.10\times$ of Section 5.2 stands for the retained term, one of two.

5.7 LIMITATIONS AND SCOPE

The three panels sit on one Qwen3.5/H20 stack, none replicates another, none is pooled, and they support bounded point estimates, not broad preservation. The archival run replays from raw outcomes but is not source-frozen regeneration: its launcher used a mutable code directory, froze no full-source or model-weight ledger, and left parts of its toolchain unarchived (Appendix A); it has no timing, and no cohort measured Recall. H20 is not an edge device, so edge use is motivation, not an evaluated claim; the calibrated per-layer row is one operating point from four disjoint items; the split depth is validated against one alternative ($j = 13$), not swept; TTFT is higher than full-prefix reuse and, on the operating-point run, than dense recomputation and the published prefix caches of Section 5.3; the timing panels are three repeats whose per-repeat maxima reach 404.01 ms TPOT against a 53.45 ms median and whose generation stops after a median of four to five tokens against a 32-token cap, so no timing difference is resolved and the eight-item throughput column reconciles with no generated-token count; that panel's dense TPOT of 648.75 ms equals its own TTFT to three digits, the re-prefill signature, against 53.46 ms for the operating-point run's prefill-once dense arm, a discrepancy we flag and do not reconcile, its shards not being re-analysed; and we do not evaluate continuous batching, QPS, admission, eviction, or scheduler effects.

Store, the deduplicated sum of the tensor-storage bytes one entry owns, excludes non-tensor/Python metadata, allocator pools, active workspace, process/NVML memory, and weights; reducing it does not by itself prove lower peak VRAM or serving capacity, and the $54.5\%/42.2\%$ allocator result is a post-priming delta against a full-copy control, neither additive with Store nor evidence of higher admitted capacity. The per-request transient is now measured at $N \leq 8$ (Section 5.6) and is higher for Q-COMEM than for the exact cache throughout that range; it is a CUDA-allocator peak, not process or NVML memory, and neither it nor Store fixes a concurrency limit for either arm, since we evaluate no admission or eviction policy.

The primary FORKAUDIT factorial has not been run on the quantized Read path: every one of its ownership cells uses full-prefix BF16 KV with no split depth or dequantization, whereas Tables 4 and 22 measure a $j = 7$ packed Q4/Q4/Q8 path whose Read step materializes a full private copy rather than borrowing it. The composition run of Section 5.6 closes that gap only on its own terms: one entry, one depth, one frozen bit vector, one Transformers stack, eight items, $N \leq 8$, and a Read path built to borrow rather than the one those tables ran, to which it is token-identical but not identical in implementation — so ownership on the exact code path behind those tables still transfers by design argument, and that run establishes nothing about paged kernels, vLLM or SGLang, a second backbone, throughput, admitted capacity, or security. Within the TCB of Section 4.4 FORKAUDIT upgrades semantic equality to phase-indexed ownership evidence, but its selected-cell binding is a same-process check: other cells and coordinates, framework/IPC semantics, compiled GDN, ATen/CUDA, and device/driver execution stay unattested, and constructed controls plus one historical mechanism show fixed-case sensitivity, not population error rates (Appendix I).

6 CONCLUSION

We report total memory, $M_{\text{total}}(D, N) = S\,D + T(N)$: the retained term alone is half the budget. The frozen Q4/Q4/Q8 point cuts S by $10.9965\times$ ($14.10\times$ native) at -0.45 $[-2.06, 0.99]$ mean F1 and raises T above the exact cache's at every fanout measured, so on the sum it wins above roughly five warm documents at low concurrency and two at $N = 8$ and loses below, under three assumptions we state rather than hide. A composition run shares one packed entry across those fanouts, passing all ten contract targets at token identity with the published path, at 4% shared. Costs stay explicit — exact reuse, dense recomputation and published prefix caches all reach the first token sooner and throughput roughly halves, so no latency result is supported — and the claim is capacity-first: smaller total memory for a large warm set.

ETHICS STATEMENT

This work introduces no human-subject data collection. It uses public PG-19 text for the trace-validation protocols and a fixed 60-item Qasper/2WikiMQA cohort, whose eight-item subset carries the latency measurements. It does not assess model safety, bias, privacy, or downstream deployment quality. Memory efficiency can lower deployment cost but may also increase the scale of model use; those broader effects are outside this controlled systems study.

USE OF LARGE LANGUAGE MODELS

The authors used LLM-based assistance throughout: drafting and revising prose, LaTeX, and tables; triaging related work; generating internal critiques and revision plans that the authors adjudicated; and drafting experiment specifications, including the registered response plan that selected which additional runs this revision executed. Every experiment ran on author-controlled infrastructure, and the authors verified each numerical claim against the archived artifacts and take full responsibility for the content, including any remaining errors.

REPRODUCIBILITY STATEMENT

Claims map to registered evidence identifiers. The 60-item table maps to a raw-first archival package that rechecks 66 mirrored files, 48 shards, 360 item-level F1 values, 24 paired-bootstrap intervals, and Store accounting; the 18 further intervals of Appendix B are recomputed from those same archived per-item values, and the operating-point run's ten pairwise intervals from its own archived rows at seed 20260903. Table 4's operating-point block comes from a separate registered execution (one warm-up, three repeats, seed 20260903) under its own run identifier, not part of that package and never pooled with it. The composition run of Section 5.6 is a further registered execution under its own identifier — a per-rank preflight gate, then a formal run whose shards carry per-target coverage and predicate rows, ownership byte accounting, and token traces — aggregated by a replay that re-derives every target status from the row's own coverage and predicate and reports any drift as a defect; it is pooled with no other panel. The latency table is hash-bound to its eight workload shards and aggregate, and manifest-first replay covers the primary ownership factorial and named supporting cohorts. The research package accompanies this submission as anonymous supplementary material, verified by content hash; it archives the quantizer/split-replay module and the prompt-and-scoring driver byte-identically, but not the panel runner, the launcher, the aggregation and bootstrap scripts, the layer-policy search, or the deployment accountant, which must be released before Table 4 is independently re-executable. That table is therefore replayable from archived outcomes and not re-executable today. Appendix A reports replay counts, costs, non-reexecuted boundaries, and the unarchived parts of the 60-item toolchain.

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A Detailed Reproducibility and Artifact Scope

The 60-item quality package (archive alias QP-60; manifest SHA-256 `24ac6952c6235bca6342171a9769f9da2d0e50d021e32c6ed453d931fd6c9952`) preserves 66 authoritative mirrored files from its recorded execution, including 48 configuration-by-rank shards. Its raw-first verifier recomputes all 360 item-level F1 values, 24 paired-bootstrap intervals, Store component accounting, and the archived aggregate. The data hash and policy receipt are bound and calibration precedes validation, but the original execution did not freeze a complete source or model-weight ledger; offline outcome replay therefore does not imply source-frozen GPU regeneration. The package archives the quantizer/split-replay module and the prompt-and-scoring driver byte-identically at four paths, but not the run driver, the aggregation and bootstrap scripts, the layer-policy search, the deployment accountant, or the launch script; an artifact reviewer can therefore read the code those files call but not the files that produced the table, and they must be released before Table 4 is independently re-executable. The cohort is indexed against the LongBench-v1 release (Bai et al., 2024c), and F1 is our reimplementation of the LongBench-v1 `qa_f1_score` protocol rather than the upstream module: the citation (Bai et al., 2024b) is the protocol source, not the code that produced any number in this paper, and the two behavioural deviations (empty-prediction/empty-reference handling, ASCII-only punctuation stripping) are stated in Section 5.1. Over-budget contexts keep their first and last halves around the prompt's context marker, with a 256-token minimum. The archived receipt binds the prepared data file by content hash, but the upstream dataset revision and evaluation-code commit were not recorded at execution time and cannot be reconstructed after the fact; a re-executor must therefore match the archived data hash rather than resolve an upstream revision.

What the seven targets establish. The primary rerun covers the 96 frozen configurations of Table 6 on one Qwen3.5-35B-A3B stack (Qwen Team, 2026a;b) and eight PG-19 books (Rae et al., 2020). Every registered semantic observable matches exactly across the 96 configurations, and completed requests in both setup arms rebind at the registered boundary to request-private tensors disjoint from the bases and from completed peers. The live-binding rerun's stronger owner-row binding covers only the selected $N = 8$ shared-document/materialized-GDN cell, leaving other cells and the honest-process assumption outside it (Table 7).

What a reader receives, and what is missing. The research package accompanies this submission as anonymous supplementary material; every alias in Appendix J resolves inside it and is verified by content hash. Eleven components of the deployment stack are recorded in the accompanying method-provenance table as *not archived* in any package: the split depth, all three bit policies and the layer-policy search, the group-size verification, the 60-item panel runner and its launcher, the Store accountant, the deployment accountant, the greedy/timing driver, and the bootstrap and aggregation scripts. Those are precisely the components that select the reported operating point and produce the reported uncertainty, so Tables 4 and 22 are replayable from archived outcomes but not re-executable from the package, and we state that here rather than only in a provenance file.

Manifest-first CPU replay covers the primary-factorial package and named replayable supporting cohorts, not every contextual table. The primary package rechecks 536 artifacts, 96 cells, 288 cross-N comparisons, eight attention rows, and nine fault pairs. The target-gate-suppression package rechecks 18 cases and 24 FP32 sidecars; its disclosed wrapper changes only an inherited execution-identifier source and preserves every candidate byte. The independent-census package rechecks six IPC captures, 1,080 receiver rows, 96,660 relations, and omission/duplication/relabel controls against a preproducer 180-slot expectation. The dual-producer package binds two fresh serial executions of the same frozen producer implementation and four distinct observers, and rechecks exact zero-tolerance equality of the 1,080 observations and 96,660 relation labels per execution. The bounded two-stream replay recomputes four full-logit comparisons and both interval overlaps.

The fresh compiled-dispatch rerun binds a frozen source archive to a passing eight-rank formal aggregate, eight rank receipts, 209,920 attention calls, 635,520 GDN calls, and 224 rejected post-run negative-control decisions. The local mirror preserves the primary and formal aggregates, runtime preflight, raw/scientific ledgers, and both terminal ledgers; it does not duplicate the model view or every raw shard. The receipt closes target 5 only at the declared honest-process fixed-stack boundary: normal launcher return is not a device-side completion attestation, and compiled GDN, underlying ATen/CUDA identity, and driver/device binaries remain outside scope.

The separate live-binding rerun binds a different frozen archive to eight scientifically valid rank shards and a terminally complete formal tree. Its selected-cell hook validates 144 selected rows, 12,960 full live-storage rows, 3,840 direct clone edges, and 24 stable phase artifacts; a separately enumerated schedule counter sees all 96 clean-memory calls with zero selected-context overlaps. The post-run audit also rejects 184 formal controls and verifies 31 bytecode files, 13 descendant directories, and the 1,367-node final tree. This supports only the selected $N = 8$ shared-document/materialized-GDN cell and six preregistered owner-addressed rows per phase under an honest same-process runtime.

Within each selected witness call, the verifier (1) enumerates 60 persistent sources before the builder, (2) records the actual builder's persistent-rooted Torch-dispatch clone edges, (3) exhaustively matches 480 destinations and passes a one-use capability bound to the exact group and verifier, (4) rereads all 540 live tensors against the production serializer at three phases, and (5) publishes hash-bound phase paths for terminal reread. Any count, object, descriptor, lineage, phase-order, or path mismatch aborts before acceptance. Trusted components are hook placement, Python/Torch-dispatch and tensor/storage semantics, and honest process scheduling; group identity, object continuity, lineage cardinality, serializer fields, and publication paths are checked.

The designer–executor-separated package binds the PDF-only author freeze, a lifecycle-amended executor, five clean/mutant pairs, detached clean and pair replays, and 208 terminal files. All five clean gates and expected-first classifications pass; the pre-injection amendment was frozen before all five ranks were rerun in a new output root.

The historical-regression package binds the pre-discovery chronology, frozen pre-fix/repair sources, outcome-blind protocol and resource amendment, eight three-lane rank artifacts, 24 FP32-logit sidecars, and 24 fresh nonces. A fresh candidate-import-free replay of every rank and the aggregate is byte-identical to the archived output.

The expanded numerical-sweep package binds 272 sidecars totaling 140,199,936 bytes and rechecks 20 attention and 24 GDN clean rows plus 44 controls. The Falcon package compares candidate arms against a separately executed official Transformers reference on 8 ranks and detached-replays the aggregate. HYPIC binds 4,404 timing artifacts and hash-replays all 16 Store cells. No package enumerates OS/driver allocations; every conclusion remains bounded to its declared configuration. Appendix J maps exact artifacts and boundaries.

On one Apple-silicon (M4-class) laptop CPU host, three consecutive fixed-order replays without a cache flush took a median 100.146 s (99.248–110.526 s) for the core package and 105.717 s (104.846–119.566 s) for the serial profile that additionally ran five locally complete supporting verifiers; the paired within-profile increment was 5.598 s (5.571–9.039 s). The canonical core distribution has 630 files and 892,284,156 logical bytes (850.95 MiB); 536 raw traces occupy 888,785,811 bytes, or 99.62% of manifest-payload bytes. These are local CPU replay and logical-storage measurements, not H20 capture cost, GPU perturbation, cold-start latency, online inference cost, or a matched uninstrumented delta.

B Complete Interval Inventory and the Reference-Arm Change

Every paired-bootstrap interval that exists for the 60-item panel is listed here, so that none is computed but unreported. The archival analysis computed 24 intervals: six configurations against dense and against split, native dtype pooled (12), and six configurations against dense on each dataset (12). *None used the native-dtype full prefix as its reference arm*, although the Store column and every compression ratio in Table 4 do. This revision therefore recomputes 18 further intervals against the native-dtype full prefix — six configurations $\times$ pooled/Qasper/2WikiMQA — from the same 360 archived item-level F1 values, and Table 4 now displays the six pooled ones. It also adds all ten pairwise comparisons among the operating-point run's five arms, the block that previously reported only differences against its own full-prefix reference. No measurement was repeated: the recomputation reproduces all six published mean-F1 values and all five published Store values exactly (the largest deviation of any recomputed point estimate from the archived one is 6.11×10^{-16} F1 points), and reproduces the archived intervals to within Monte-Carlo noise. The archived analysis records no bootstrap seed, so its endpoints are reproducible only up to that noise; the new intervals use seed 20260902, 10,000 paired item-level resamples of the 60 common $(\text{dataset}, \text{source index})$ keys, and a percentile interval. Because seed plus resample count does not fix percentile endpoints across RNG libraries, we state the generator: a Python `random.Random` seeded once per comparison, drawing resample indices one at a time in $\text{randrange}(n)$ order, with the reported endpoints taken as the order statistics at indices $\lfloor 0.025(n-1) \rfloor$ and $\lfloor 0.975(n-1) \rfloor$ of the sorted resampled means. Endpoints reproduce exactly only for this generator, seed sequence, and repetition count. Across five seeds the frozen row's endpoints move by at most 0.075 points, so the intervals are quoted to two decimals.

Two consequences of the change of arm are worth stating. First, the two steps compound rather than cancel: -0.0585 for splitting plus -0.3870 for quantizing the split state equals the -0.4455 reported against full-prefix, so the previously published -0.39 understated the deployment-relevant loss by about 13% of its value. Second, the frozen policy's registered safety count is unchanged by the change of arm: $0/60$ items below the -50-point catastrophic threshold against dense and $0/60$ against full-prefix.

The ten operating-point pairs come from a separate pass over that run's 60 common item keys, 10,000 paired item-level resamples at seed 20260903, and the same generator and percentile convention. Their point estimates reproduce the four differences Table 4 prints against the full-prefix arm exactly, and their endpoints agree with the printed ones to within 0.20 F1 points — the resampling noise of an independently drawn resample sequence, of the same order as the 0.075-point five-seed movement recorded above. One pair, full-prefix Q8 minus frozen $j = 7$ at -3.4122 $[-8.4296, -0.0390]$, excludes zero; the other nine do not. That one is the comparison earlier revisions declined to compute, and its upper endpoint lies 0.0390 points from zero, so it supports the ordering and nothing about the size or the robustness of the effect. None of the ten carries a multiplicity adjustment, and none is an equivalence or noninferiority test. Two ablation details from Section 5.4 belong with this inventory: on the 60-item panel the calibrated per-layer policy saves 2.75% Store while its mean-F1 difference is more negative than the frozen point's and one of 60 items crosses the catastrophic threshold, and uniform Q4 loses 3.047 points on the eight-item execution.

Why dense is not the per-item reference. The archived rows show identical prefix, document, query and input token counts, identical truncation flags, and identical items, questions and references across all six arms, so the wide dense interval is not an execution-boundary artifact. What the rows do show is that dense recomputation and cached-prefix continuation emit different greedy token sequences on 6 of 60 items, two of them short-answer 2WikiMQA questions whose per-item F1 is nearly binary; the per-item difference standard deviation is 14.51 against 0.45 for full-prefix versus split at native dtype, which is why the interval is $41\times$ wider. Logits are not archived, so we report the divergence rather than asserting its mechanism. By the same measure Q-CoMem split replay at native dtype tracks the full-prefix reference more closely ($58/60$ identical token sequences) than dense recomputation does ($54/60$), and on both items where it differs its output is the one that matches dense. The dense row is therefore retained as a no-retention reference, not as a per-item baseline.

Table 2: All 52 paired-bootstrap intervals for the 60-item panels, on the 0–100 F1 scale. The first block is the 18 intervals computed against the native-dtype full prefix on the archival run (seed 20260902); the second is the operating-point run's ten pairwise comparisons (seed 20260903), which complete that block's inventory; the third is the 24 intervals in the archival analysis, with the reference arm each one actually used. Rows in the first block with a pooled subset are the ones displayed in Table 4.

Configuration	Subset	ΔF1	95% CI
This revision; reference arm: full-prefix, native dtype (18 intervals)			
No-cache dense recompute	pooled	-0.3934	$[-4.6648, +2.6049]$
No-cache dense recompute	Qasper	$+1.2131$	$[-0.2339, +3.3305]$
No-cache dense recompute	2WikiMQA	-2.0000	$[-10.0000, +4.0000]$
Full-prefix, native dtype	pooled	$+0.0000$	$[+0.0000, +0.0000]$
Full-prefix, native dtype	Qasper	$+0.0000$	$[+0.0000, +0.0000]$
Full-prefix, native dtype	2WikiMQA	$+0.0000$	$[+0.0000, +0.0000]$
Q-CoMem split, native dtype	pooled	-0.0585	$[-0.1754, +0.0000]$
Q-CoMem split, native dtype	Qasper	-0.1170	$[-0.3509, +0.0000]$
Q-CoMem split, native dtype	2WikiMQA	$+0.0000$	$[+0.0000, +0.0000]$
Q-CoMem frozen Q4/Q4/Q8	pooled	-0.4455	$[-2.0586, +0.9907]$
Q-CoMem frozen Q4/Q4/Q8	Qasper	-0.3354	$[-3.3818, +2.3932]$
Q-CoMem frozen Q4/Q4/Q8	2WikiMQA	-0.5556	$[-1.6667, +0.0000]$
Q-CoMem calibrated per-layer	pooled	-1.3174	$[-4.3322, +0.9193]$
Q-CoMem calibrated per-layer	Qasper	-0.2539	$[-3.3148, +2.3720]$
Q-CoMem calibrated per-layer	2WikiMQA	-2.3810	$[-7.1429, +0.0000]$
Q-CoMem aggressive mixed	pooled	-5.4961	$[-11.5805, -0.1984]$
Q-CoMem aggressive mixed	Qasper	-3.9445	$[-11.4709, +1.5691]$
Q-CoMem aggressive mixed	2WikiMQA	-7.0476	$[-17.7143, +1.3333]$
Operating-point run; all ten pairwise comparisons, pooled (seed 20260903)			
Dense − full-prefix, native dtype	pooled	-0.4881	$[-4.5660, +2.5509]$
Dense − full-prefix Q8	pooled	$+3.2949$	$[-2.4786, +9.4520]$
Dense − frozen Q4/Q4/Q8, $j = 13$	pooled	$+2.9071$	$[-2.8268, +8.9943]$
Dense − frozen Q4/Q4/Q8, $j = 7$	pooled	-0.1173	$[-4.4576, +3.1106]$
Full-prefix, native dtype − full-prefix Q8	pooled	$+3.7831$	$[-0.1914, +9.3195]$
Full-prefix, native dtype − frozen, $j = 13$	pooled	$+3.3952$	$[-0.5737, +8.7639]$
Full-prefix, native dtype − frozen, $j = 7$	pooled	$+0.3708$	$[-1.0533, +2.0345]$
Full-prefix Q8 − frozen Q4/Q4/Q8, $j = 13$	pooled	-0.3878	$[-5.3363, +4.6122]$
Full-prefix Q8 − frozen Q4/Q4/Q8, $j = 7$	pooled	$\mathbf{-3.4122}$	$[\mathbf{-8.4296}, \mathbf{-0.0390}]$
Frozen $j = 13$ − frozen $j = 7$	pooled	-3.0244	$[-8.2344, +0.5789]$
Archival analysis; reference arm as listed (24 intervals)			

Continued on next page

Table 2 continued.

Configuration	Subset	ΔF1	95% CI
Configuration	Reference arm	Subset	95% CI
No-cache dense recompute	dense	pooled	[+0.0000, +0.0000]
No-cache dense recompute	split, native dtype	pooled	[−4.6063, +2.6650]
No-cache dense recompute	dense	2WikiMQA	[+0.0000, +0.0000]
No-cache dense recompute	dense	Qasper	[+0.0000, +0.0000]
Full-prefix, native dtype	dense	pooled	[−2.6066, +4.5088]
Full-prefix, native dtype	split, native dtype	pooled	[+0.0000, +0.1754]
Full-prefix, native dtype	dense	2WikiMQA	[−4.0000, +10.0000]
Full-prefix, native dtype	dense	Qasper	[−3.3305, +0.2339]
Q-CoMem split, native dtype	dense	pooled	[−2.6650, +4.4572]
Q-CoMem split, native dtype	split, native dtype	pooled	[+0.0000, +0.0000]
Q-CoMem split, native dtype	dense	2WikiMQA	[−4.0000, +10.0000]
Q-CoMem split, native dtype	dense	Qasper	[−3.4475, +0.0000]
Q-CoMem frozen Q4/Q4/Q8	dense	pooled	[−3.3045, +4.2800]
Q-CoMem frozen Q4/Q4/Q8	split, native dtype	pooled	[−2.0365, +1.0756]
Q-CoMem frozen Q4/Q4/Q8	dense	2WikiMQA	[−4.3333, +9.4444]
Q-CoMem frozen Q4/Q4/Q8	dense	Qasper	[−4.6073, +0.7317]
Q-CoMem calibrated per-layer	dense	pooled	[−5.2845, +3.7838]
Q-CoMem calibrated per-layer	split, native dtype	pooled	[−4.2793, +0.9775]
Q-CoMem calibrated per-layer	dense	2WikiMQA	[−8.4762, +8.6667]
Q-CoMem calibrated per-layer	dense	Qasper	[−4.9649, +1.6662]
Q-CoMem aggressive mixed	dense	pooled	[−11.9941, +1.5786]
Q-CoMem aggressive mixed	split, native dtype	pooled	[−11.5615, −0.1713]
Q-CoMem aggressive mixed	dense	2WikiMQA	[−16.7619, +6.6667]
Q-CoMem aggressive mixed	dense	Qasper	[−12.9395, +1.0689]

C Per-Component Store Accounting and the Format Ceiling

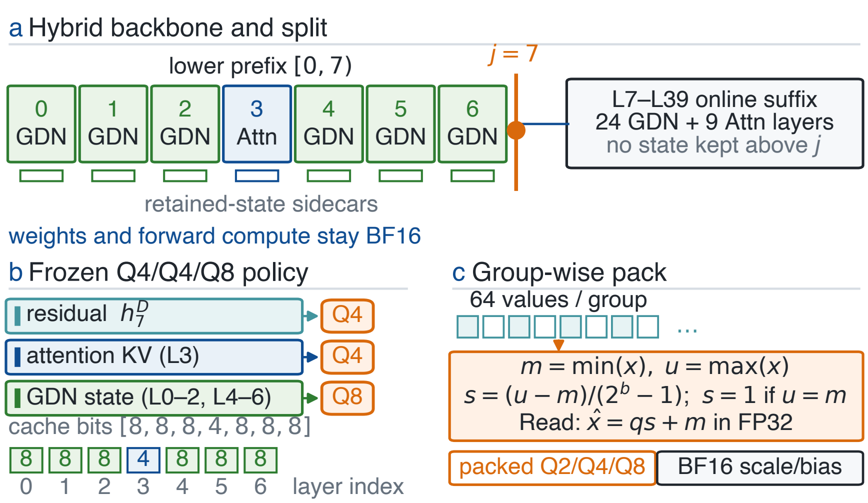

Figure 2: Quantization scope and the frozen Q4/Q4/Q8 policy. (a) At split $j = 7$ the retained entry holds the boundary residual and the lower-layer (L0–L6) state sidecars; weights and forward computation stay BF16, and L7–L39 state is reconstructed on Read. (b) The frozen policy's cache bit vector is $[8, 8, 8, 4, 8, 8, 8]$ with a Q4 residual: h_7^D and the L3 attention KV at Q4, the L0–L2/L4–L6 GDN convolution and recurrent state at Q8. The calibrated per-layer policy is the different vector $[8, 8, 4, 4, 8, 8, 8]$ and is not drawn here. (c) Each 64-value group stores packed unsigned codes plus a BF16 scale and bias. The unpacked reference is the entry at native dtype — BF16 for the residual and lower attention KV, the stack's FP32 for GDN recurrent state — and we do not call it "Q16". The scope is retained tensor payload, not weights or total VRAM.

In both deployment tables the residual term of “Store” is $\mathrm{numel}\times$ element size and the cache term sums untyped-storage bytes deduplicated by (device, data pointer, storage size). On the eight documents both deployment tables contain, the printed deduplicated sum and the entry’s physical byte-range union agree to the byte at all five retained configurations, so the Section 5.1 correction moves no number.

Table 4’s Store column is a sum of per-component byte counts, and this appendix prints them so that each row can be checked against Eq. 3 without back-derivation. For a component of n values packed at $b < 16$ bits in groups of 64 with BF16 scale and bias, the packed size is

$$\mathrm{bytes}(n, b) = \lceil n/64 \rceil \, (8b + 4),$$

so the format ceilings against a 128-byte BF16 group are $3.5556\times$ at $b = 4$ and $1.8824\times$ at $b = 8$. Every quantized item–policy pair in the archival cohort satisfies this identity exactly (180/180, zero mismatches).

Three details Eq. 3 omits are needed to reimplement it: groups are taken inside the hidden dimension for the residual (requiring hidden size divisible by 64) and over the flattened element order with edge-value padding for the cache leaves; a constant group would make Eq. 3 degenerate, so the implementation sets $s = 1$ when $u = m$; and Read reconstructs $\hat{x} = qs + m$ in FP32 from the stored BF16 s and m, rounded to BF16 after q is computed, then casts back to the leaf’s original dtype, so an FP32 leaf round-trips as FP32.

Table 3 gives each component at the cohort mean document length $T = 3807.25$ tokens. Two columns are needed for the unpacked reference: the stack retains Qwen3.5 GDN recurrent state in FP32, so its *native* cost is twice the cost it would carry under an all-BF16 encoding. Every other component is BF16 natively, and $j = 7$ places one full-attention layer (layer 3) and six GDN layers below the split.

Table 3: Per-component retained payload in MiB/document at the 60-item mean document length ($T = 3807.25$ tokens), by stored width. “Native” is the dtype the unmodified stack produces; “BF16 ref.” re-expresses GDN recurrent state in BF16. Quantized entries are $\lceil n/64 \rceil (8b + 4)$ bytes.

Component	values n	Native	BF16 ref.	Q8	Q4	Q2
Boundary residual h_j^D	$2048\,T$	14.8721	14.8721	7.9008	4.1828	2.3238
Layer-3 attention K	$512\,T$	3.7180	3.7180	1.9752	1.0457	0.5809
Layer-3 attention V	$512\,T$	3.7180	3.7180	1.9752	1.0457	0.5809
GDN convolution state, per layer	32,768	0.0625	0.0625	0.0332	0.0176	0.0098
GDN recurrent state, per layer	524,288	2.0000	1.0000	0.5312	0.2812	0.1562

Table 4: Retained-state–quality trade-off on the 60-item validation cohort (30 Qasper, 30 2WikiMQA) in two separate executions, blocked and never pooled, each differenced against its own full-prefix row. The all-BF16 columns come first because they isolate packing from the stack's dtype choices.

Retained state	Store, BF16 ref. MiB/doc	vs.	Store, native MiB/doc	vs.	TTFT (s)	F1 (0–100)	ΔF1 vs. full-prefix [95% CI]
Archival quality run (seed 20260902; no timing recorded)							
No-cache dense recompute	–	–	–	–		54.29	-0.39 [-4.66, 2.60]
Full-prefix, native dtype	106.235	1.00×	136.235	1.00×		54.68	0.00 [0.00, 0.00]
Q-CoMem split, native dtype	28.683	3.7038×	34.683	3.93×		54.62	-0.06 [-0.18, 0.00]
Q-CoMem frozen Q4/Q4/Q8	9.661	10.9965×	9.661	14.10×		54.24	-0.45 [-2.06, 0.99]
Q-CoMem calibrated per-layer	9.395	11.3073×	9.395	14.50×		53.36	-1.32 [-4.33, 0.92]
Q-CoMem aggressive mixed	7.536	14.0969×	7.536	18.08×		49.19	-5.50 [-11.58, -0.20]
Operating-point run (seed 20260903; separate execution on the same 60 items)							
No-cache dense recompute	0.000	–	0.000	–	0.636	54.18	-0.49 [-4.57, 2.55]
Full-prefix, native dtype	106.235	1.00×	136.235	1.00×	0.181	54.67	0.00 [0.00, 0.00]
Full-prefix Q8 (uniform 8-bit)	56.438	1.8823×	56.438	2.41×	0.182	50.89	-3.78 [-9.12, 0.32]
Q-CoMem frozen Q4/Q4/Q8, $j = 13$	16.101	6.5980×	16.101	8.46×	0.582	51.27	-3.40 [-8.67, 0.52]
Q-CoMem frozen Q4/Q4/Q8, $j = 7$	9.661	10.9965×	9.661	14.10×	0.656	54.30	-0.37 [-1.97, 1.08]

Store and F1 are 60-item means; each block uses its own full-prefix arm. "Native" keeps GDN recurrent state in FP32, "BF16 ref." re-expresses it (Appendix C). Full-prefix Q8 packs the unsplit entry at a uniform 8-bit width, *not* the frozen policy's bit vector, so the $5.84\times$ between it and the $j = 7$ row is width-confounded (Section 5.2). Intervals are paired 10,000-resample item-level bootstrap percentile intervals against that block's full-prefix arm (seeds 20260902/20260903) on unrounded means; all ten pairwise comparisons among the operating-point block's five arms, the frozen-versus-Q8 pair included, are listed in Appendix B. An interval crossing zero is not statistical equivalence, the headline policy was selected without multiplicity adjustment, and the ten pairs carry none either. Recall was measured in no cohort; blank TTFT cells mark a metric the archival run did not record.

Each Table 4 row is a sum of these entries. The full-prefix reference spans all ten attention layers and all thirty GDN layers, $10 \times 2 \times 3.7180 + 30 \times 0.0625 + 30 \times 2.0000 = 136.235$, of which $30 \times 1.0000 = 30.000$ MiB is the FP32 excess; its all-BF16 value is 106.235. The same reference reads 140.34 MiB on the eight-document timing panel, 42.8% of it FP32 GDN recurrent state — a total, where the 30.000 MiB here is an excess over BF16, and not two readings of one quantity. That 136.235-versus-140.34 gap is cohort and statistic only: the operating-point run's document token counts are identical across its five arms (minimum $1{,}146$, median $4{,}000$, maximum $4{,}050$), its own full-prefix mean Store is 136.2354 MiB/document, matching the archival cohort exactly, and the archived per-item rows attribute 69.3% of the gap to which eight documents the timing panel draws and 30.7% to mean versus median, 0.0% to a difference of estimand. Split, native dtype, retains the residual, layer-3 K and V, and six GDN layers, $14.8721 + 2 \times 3.7180 + 6 \times 0.0625 + 6 \times 2.0000 = 34.683$, of which $6 \times 1.0000 = 6.000$ MiB is the same excess, so its all-BF16 value is 28.683. The frozen policy packs the residual and layer-3 K/V at Q4 and the six GDN layers at Q8, $4.1828 + 2 \times 1.0457 + 6 \times (0.0332 + 0.5312) = 9.661$; the calibrated per-layer policy moves layer 2 to Q4, giving 9.395; and the aggressive policy uses Q2 for layer-3 attention and for GDN layers 2, 4 and 6, giving 7.536. The calibrated per-layer policy was selected on a disjoint four-item calibration set under a byte budget set by the frozen policy's predicted size, and the aggressive policy under a budget 25% below it; both were frozen before validation.

The transient term: the arithmetic, and the measurement refuting it. Because M_{active} is method-dependent here (Section 3), we report it rather than exclude it. The component inventory predicts it as follows. The implemented Read materializes the split entry privately per concurrent request, at the cohort mean length the all-BF16 split-entry cost of 28.683 MiB/request; the full-prefix reference shares its document KV but still carries its own mutable GDN convolution and recurrent state per request, 31.875 MiB/request in BF16 and 61.875 in the stack's FP32. For N concurrent requests on one hot document the all-BF16 resident-byte lines are $9.661 + 28.683N$ against $106.235 + 31.875N$, giving Q-CoMem both the smaller intercept and the smaller slope and therefore no crossing at any N. *Measurement refutes that prediction*, and we retain it here only because we published it: the composition run (Section 5.6, Table 5) measures peak transient allocation directly and finds Q-CoMem above full-prefix at every fanout run, with the smaller slope but an intercept roughly eight times larger and a fitted crossing at $N = 12.5$. The arithmetic counts only entry-owned tensor payload; the measurement counts every allocation the query makes, including the dequantization and concatenation buffers the arithmetic never sees. Neither counts pools, fragmentation, or non-tensor metadata, and the allocator result in Section 5.5 is not additive with Store. Table 1 then sums the two measured terms into the total-memory model Section 5.6 reports, under the three assumptions stated there.

Table 5: Median peak transient allocation per query in the composition run — the CUDA-allocator peak above the pre-fork baseline, not process or NVML memory — for one packed depth-7 Q4/Q4/Q8 entry serving N concurrent requests, against the full-prefix exact cache on the same eight items, stack and protocol (Section 5.6).

Arm (MiB)	$N=1$	$N=2$	$N=4$	$N=8$
Full-prefix exact cache	286.44	427.49	752.09	1441.93
Q-COMEM packed $j=7$, one shared entry	930.91	1062.73	1178.41	1713.15
Q-COMEM packed $j=7$, private per request	954.09	1063.35	1393.28	1976.98
Least-squares fits: full-prefix $103.45+166.28N$; shared $809.68+109.76N$; private $790.50+148.38N$				

The exact-cache arm, and the width-matched point it is not. The same identities rebuild the operating-point run's quantized exact cache and expose the bit-width confound in the $5.84\times$ that Section 5.2 reports. Full-prefix Q8 packs all twenty attention K/V components and all thirty GDN layers at Q8: $20\times1.9752+30\times(0.0332+0.5312)=39.504+30\times0.5644=56.436$ MiB/document, reproducing the printed 56.438 to 0.002 MiB — the rounding of the per-component entries in Table 3. Applying the frozen policy's *own* widths to the same unsplit entry — Q4 on attention, Q8 on GDN — gives $20\times1.0457+16.932=20.914+16.932=37.846$ MiB/document. The measured ratio therefore factors exactly:

$$\underbrace{\frac{56.436}{37.846}}_{1.4912\text{ (bit width)}} \times \underbrace{\frac{37.846}{9.661}}_{3.9174\text{ (split depth)}} = 5.8416 = \frac{56.436}{9.661}.$$

Only the 3.9174 is attributable to splitting, and it agrees with the $3.93\times$ split step the archival panel measures against the native-dtype reference. *The* 37.846 *MiB/document arm was never executed.* It is a byte count fixed by the format identity above once the bit vector is given, in exactly the sense that the all-BF16 column is a re-expression rather than a run; it carries no F1, no interval, and no claim about whether the split keeps its quality advantage at matched precision. Establishing that requires the arm.

This is what dissolves the apparent format violation in the printed column. Taken as printed, the quantization step reads $34.683/9.661=3.590\times$, above the $3.5556\times$ Q4 ceiling. Both numerator and denominator must be stated under one dtype convention: with the split row's 6.000 MiB of FP32 excess removed, the step is $28.683/9.661=2.969\times$, inside the ceiling, and the lower-cache-only sub-ratios are $2.5211\times$ (frozen) and $2.6496\times$ (calibrated per-layer) against the same $3.5556\times$ bound, with the aggressive policy at $4.1187\times$ against its own $6.4000\times$ Q2 bound. The printed $3.590\times$ is a physical saving on this stack that mixes packing with dtype narrowing; it is not a claim that Eq. 3 compresses beyond its format.

D PROTOCOL GEOMETRY

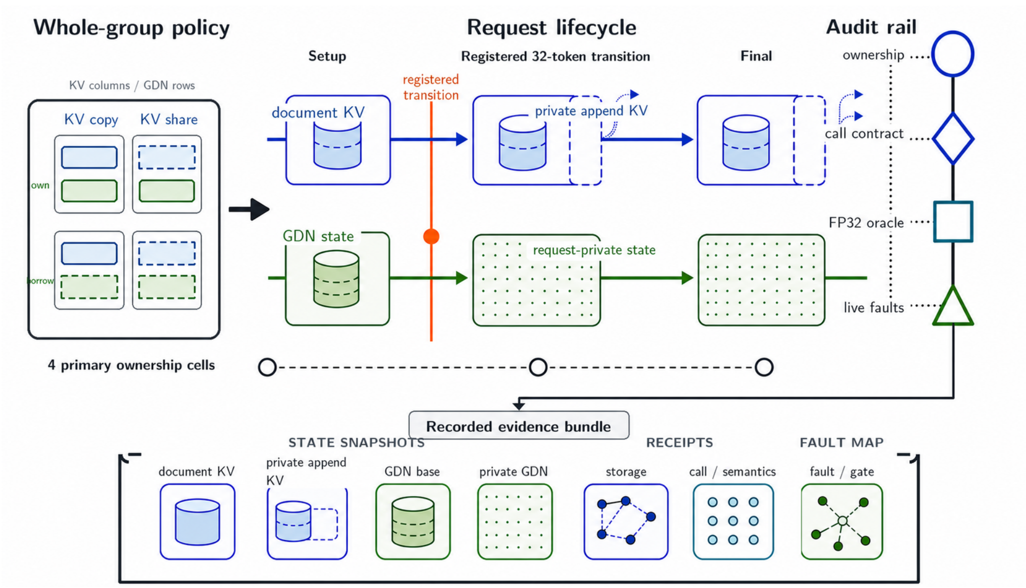

Figure 3: FORKAUDIT ownership factorial. Four KV-by-GDN setup cells share immutable document state, keep append tails private, and rebind borrowed recurrent state at the registered transition. The audit rail records lifecycle, storage, call/semantic, and fault-to-gate receipts. This schematic details the validation component; it is not experimental evidence.

Table 6: Frozen primary-factorial execution geometry. The eight ranks provide distinct books and query banks, not repeated stochastic measurements.

Component	Frozen value
Model	Qwen3.5-35B-A3B; 10 attention + 30 GDN layers
Hardware	8 $\times$ H20-3e; one book/rank
Software	Python 3.11.13; PyTorch 2.11.0+cu129; Transformers 5.14.1; vLLM 0.26.0+cu129; Triton 3.6.0; CUDA 12.9
KV backend	vLLM fused attention; BF16; page 128
Document / query	4,095 / 32 tokens; PG-19 train only
Generation	8 output tokens; 39 state-appended input tokens (32 query + 7 generated-token feedback)
Resident counts	$1, 8, 32$; four KV-by-GDN ownership cells
Primary schedule	all requests resident; batch-one sequential round-major calls on one CUDA stream/rank
Validation	96 factor configurations; separate memory/witness cells
Fault campaigns	$N = 2$; the primary all-gates-on campaign has nine live mutants with rebuilt controls; the target-gate-suppression campaign has nine fresh clean plus nine suppressed-gate cases on 8 H20s; the designer–executor-separated campaign has five new clean/mutant pairs on 5 H20s
Historical regression	one previously encountered one-token alias defect; 8 ranks $\times$ 3 fresh $N = 2$ lanes; 3 archived-coordinate reproductions + 5 additional frozen inputs; no mutation

Table 7: Meaning of the live-binding rerun's selected-cell evidence units (archive entry E-LIVE-BINDING-A; Appendix J). Counts are per rank unless marked total; all belong to the same formal run.

Unit	Count	Check and authorization
Persistent sources	60	independently enumerated before the actual builder; exact object, storage, descriptor, and content remain fixed
Selected anchors	6/phase	preregistered persistent/request semantic rows; request-owned subset consumes the group-bound lineage capability
Full live/serializer rows	540/phase	$(1 \text{ persistent} + 8 \text{ requests}) \times 60$ rows; live objects are re-enumerated and every serialized field, role, interval, content digest, and phase relation is checked
Builder clone edges	480	8×60 materialized destinations exhaust the captured persistent-rooted ledger and are direct aten.clone outputs
Phase artifacts	3	setup, post-transition, and post-generation products are hash-bound, published to stable rank paths, and terminally reread
Clean-memory context separation	12 (96 total)	separate allocator-cell calls are all observed; zero selected-context overlaps mean the selected context did not enter them, not that the wrapper was absent or perturbation-free

With field order (owner, request, layer, family, state), the six anchors are $(\mathrm{Pers}, \varnothing, 0, \mathrm{rec}, 0)$, $(\mathrm{Req}, 0, 0, \mathrm{conv}, 0)$, $(\mathrm{Req}, 0, 1, \mathrm{conv}, 0)$, $(\mathrm{Req}, 0, 2, \mathrm{conv}, 0)$, $(\mathrm{Req}, 0, 2, \mathrm{rec}, 0)$, and $(\mathrm{Req}, 1, 0, \mathrm{rec}, 0)$. The capability covers the five request rows; the persistent row is their source anchor.

Table 8: Cohort-to-claim authorization. Cohorts are never pooled; each row supports only the listed inference.

Cohort	Frozen scope	Authorized claim / explicit exclusion
Primary ownership factorial	8 books; PG-19; full-prefix BF16 KV; no split depth; no quantization; $N = 1, 8, 32$; four KV×GDN cells; 8 decode steps	ownership lifecycle, cross-cell equivalence, selected attention oracle, live-mutant sensitivity, and four-cell allocator endpoints; not the quantized Read path, and not transfer or scheduler evidence
Primary target-5 rerun	same frozen 8-rank, 96-configuration Qwen/H20 factorial; 209,920 attention and 635,520 GDN calls	per-call selected Triton kernel-cache artifact/configuration and eager-route binding at the declared fixed-stack scope; not device-side completion, driver/device, compiled-GDN, ATen/CUDA, performance, or cross-stack evidence
Selected-cell live binding	same frozen 8-rank factorial; stronger hook active only for the $N = 8$ shared-document/materialized-GDN cell; actual builder, production serializer, and direct clone lineage	144 selected rows, 12,960 live-storage rows, 3,840 clone edges, and 24 phase artifacts pass; all 96 clean-memory calls have zero selected-context overlaps; not evidence for other cells or coordinates, malicious runtimes, framework independence, device binaries, or performance
Fresh-process control	one independent $N = 2$ process; one clean path and M1	input reconstruction, cleanup, and one intended-gate reproduction; not pooled with the factorial
Lifecycle transfer	8 books; aligned 4,096-token prefix; $N = 4$; two cancelled slots	exact outputs/state, zero-scrub, exact reuse, and stale-lease rejection on the same adapter; not scheduler, concurrency, or performance evidence
Scheduler interleave	8 books; $N = 4$; page/prefix/tail geometries 64/3,985/17 and 128/4,033/65; cancel slot 3 after round 2	deterministic request-step interleaving, zero-scrub, epoch change, exact reuse, uninterrupted-control equality, and three frozen scheduler-fault gates; not concurrent kernels, continuous-batching performance, detection rate, or production end-to-end correctness

Continued on next page

Table 8 continued.

Cohort	Frozen scope	Authorized claim / explicit exclusion
Independent preproducer slot census	frozen Qwen/H20 geometry and schedule; 180 slots/capture; two $N = 2$ GDN-policy cells and three phases/cell	all 6 captures, 1,080 receiver rows, and 96,660 relations close; omission, duplication, and semantic-relabel controls fail closed; producer manifests/rows are not expected-set authority, but slot-ID-to-live-tensor binding, IPC, and paused capture remain trusted
Dual-producer repeat	same frozen input/protocol and producer implementation; two fresh serial executions and four distinct out-of-process observers	both executions reproduce 1,080 semantic coordinates, byte digests, stable descriptors, and 96,660 relation labels at zero tolerance; not malicious-producer resistance, independent receiver model execution, or cross-stack evidence
Falcon-H1 bounded transfer	`tiiuae/Falcon-H1-0.5B-Base`; Transformers 5.14.1/H20 naive path; $N = 1, 2$; separately executed official Transformers reference; persistent-unquantized and deep-materialized arms	8/8 ranks, 96 generated-token and 96 full-vocabulary FP32-logit comparisons, 13,824 state rows across KV key/value, convolution, and Mamba2 recurrent families, 192 ownership checks, cross-arm/cross-N relations, and registered controls pass; no performance, memory, quality, or other-stack evidence
Bounded two-stream lifecycle	one H20; one 4,033-token document; two 16-token query requests at $N = 2$; two host threads and distinct CUDA streams; pre-cancel and post-reclaim phases; cancellation after synchronization	positive call-interval overlap plus exact four lifecycle outputs, 20 logical-KV layer pairs, two GDN slot aggregates, ownership, scrub, and reuse; not simultaneous kernels, native continuous batching, in-flight cancellation, production scheduling, or performance
GDN transition oracle	1 frozen book/window; global model-layer indices 0, 10, 20, 38; 4 query tokens	clean recurrence outputs/states and four seeded wrong-transition rejections conditional on post-normalization inputs; not all GDN layers or end-to-end semantics
Expanded captured-boundary numerical sweep	2 distinct PG-19 inputs; all 10 attention layers and 12 of 30 GDN layers; 20 attention and 24 GDN clean rows plus one seeded control/row	captured-boundary numerical agreement and rejection of all 44 seeded wrong operators; not upstream-activation, independent-capture, full-model, or end-to-end validation
Target-gate suppression matrix	9 fresh clean + 9 target-suppressed cases; same fixed model/data/$N = 2$; 8 distinct H20s	per-fault comparison with redundant audit gates, exact allowlisted production assertions, and token/FP32-logit equality; M8's injected sentinel is not a detector; not a blind-fault study or rate
Designer–executor-separated faults	separate designer reads only the prior PDF; 5 new frozen $N = 2$ clean/mutant pairs; 5 H20s; all gates enabled	all 5 clean gates pass and all 5 mutants fail first at their frozen primary predicate; per-fault sensitivity only, not a detection rate, natural-defect study, or cross-stack evidence
Historical alias regression	one pre-discovery integration defect; 3 archived coordinates + 5 additional frozen inputs; fresh pre-fix, repaired, and materialized $N = 2$ one-token lanes; 8 H20s	pre-fix binding gate and base violation with output/terminal-state equality, plus storage-clean exact repair; one retrospective defect only, not prevalence, recall, statistical replication, or cross-stack evidence
Mac common-mode control	Apple M4 Pro; six documents/queries; five fresh MLX processes; vanilla dense plus Q-COMEM BF16/Q8/Q4 at depth 7	separates split-family agreement from agreement with vanilla dense; motivation only and separate from the H20 case

Continued on next page

Table 8 continued.

Cohort	Frozen scope	Authorized claim / explicit exclusion
H20 archival Store–F1 validation	60 validation items from Qasper/2WikiMQA (30 each); dense, full-prefix, split, native dtype, frozen Q4/Q4/Q8, and two mixed policies	raw-first replay of Store, predictions, mean F1, paired bootstrap intervals, and fixed-case failure counts; not source-frozen regeneration, TTFT/TPOT/throughput, Recall, total memory, edge, or broad-benchmark evidence
H20 operating-point run	the same 60 validation items; dense, full-prefix native dtype, full-prefix Q8, and Q-CoMem frozen Q4/Q4/Q8 at $j = 13$ and $j = 7$; one warm-up, three repeats, dataset-length generation, seed 20260903	Store, TTFT, TPOT, mean F1, and paired bootstrap intervals against full-prefix from a separate execution; not pooled with the archival panel, not Recall, total memory, admitted capacity, resident-set, edge, or ForkAudit evidence
Composition run (packed entry shared at $N > 1$)	the same eight-item slice (source indices 6–9); one packed $j = 7$ Q4/Q4/Q8 entry; fanouts $N \in \{1, 2, 4, 8\}$; cross-item queries; 32 generated tokens; one warm-up, three repeats, seed 20260903; shared-packed view with registered-transition rebind	shared-mode effectiveness with no fallback, non-vacuous sharing at the *final* window, per-request ownership tensor/byte split, token identity against the published $N = 1$ private-materialization path, ten-target ForkAudit coverage and verdicts, and peak transient allocation per fanout; full-prefix token agreement is a recorded diagnostic that gates nothing; not paged-kernel, vLLM/SGLang, throughput, admitted-capacity, second-backbone, quality, or security evidence
H20 deployment timing	source indices 6–9 from each of Qasper/2WikiMQA (8 items total), a subset of the 60-item validation cohort; no-cache dense recompute, full-prefix KV reuse, and Q-CoMem Q8/Q4/mixed variants (six configurations total) with 3 timing repeats	same-stack retained-Store, TTFT, TPOT, tokens/s, and mean F1 comparison from a separate execution; not an independent replication, not pooled with the 60-item panel or primary factorial, and not scheduler/continuous-batching evidence
Matched serving controls	8 validation items from Qasper/2WikiMQA; vLLM 0.26 and SGLang 0.5.17 cache off/on plus 24 fixed-configuration HYPIC timing/quality cells; one H20-3e per item	within-block cache-hit, quality, and client-wall timing context; not pooled with in-process Q-CoMem timings, not a cross-framework leaderboard, and not continuous-batching evidence
HYPIC retained-state receipts	8 Prefix Cache + 8 HYPIC cells on the same fixed TP=1 model/slice; receipt-instrumented SGLang lifecycle	median target-entry-owned physical tensor payload only; not timing, NVML/process memory, capacity, continuous batching, or ForkAudit ownership evidence
Related-work operator transfer	Hydragen at $N = 8/32$ on one captured layer-3 attention family; Palu on one layer with 8 calibration and 1 held-out PG19 row	same-checkpoint numerical and logical-state trade-offs only; not an all-layer implementation, LongBench/quality result, speedup claim, or jointly optimized system
Local replay/storage accounting	one Apple-silicon (M4-class) laptop CPU host; three consecutive fixed-order runs without cache flush; manifest-only core package plus a serial profile with five locally complete supporting verifiers	local CPU replay wall time and logical file/byte counts only; not H20 capture, GPU perturbation, cold-start latency, online inference, compressed-upload size, or an uninstrumented-baseline delta

Table 9: The seven contract targets of Section 4.4: trace coverage, replay verdict, what a pass rules out, and the execution that produced it. A pass is trace-relative and conditional on capture honesty, slot-to-live-tensor binding, and the target boundary of Section 4.4; it is not comprehensive system correctness. Appendix Table 10 gives the decisive witness for each target.

#	Target	Cov.	Verd.	A pass rules out	Execution; scope limit
1	Frozen identity	compl.	pass	a changed source, model, data, query, or protocol between capture and replay	primary rerun; one formal run
2	Prefix immutability	compl.	pass	in-place writes to shared document KV or the GDN base at any phase	primary rerun; one frozen geometry
3	Private ownership	compl.	pass	a request that still writes through an alias of the base or of a peer	primary rerun; live-binding rerun adds six owner-addressed rows per phase in one cell
4	Tail-safe append	compl.	pass	a COW-elided in-place write to the partial final page	primary rerun; one partial-tail geometry
5	Host-side launch/config.	compl.	pass	silent kernel fallback or route contamination on the host side	dispatch rerun; normal launcher return, not device completion
6	Cross-arm equivalence	compl.	pass	an ownership arm that is cheaper because it computes something different	primary rerun; one sequential schedule
7	Cross-N consistency	compl.	pass	a prefix whose reuse depends on how many requests are resident	primary rerun; relational, not an independent oracle

Table 10: Trace coverage and replay verdict for the seven contract targets in fresh frozen primary case. Equality denotes canonical byte-digest identity as recorded. Overall trace coverage is complete and all seven target verdicts pass at their declared fixed-stack scope. A passing primary target verdict remains trace-relative and conditional on capture honesty, slot-ID-to-live-tensor binding, and the target boundary in Section 4.4. The census removes producer rows as expected-set authority only for its bounded GDN cohort; supporting cohorts do not upgrade these targets to comprehensive system correctness.

Target	Trace coverage	Replay verdict	Decisive witness	Boundary
Frozen identity	complete	pass	identical preflight/terminal source ledgers, terminal model rehash, and stable data/query/protocol/execution bindings	one primary-factorial formal run; the dual repeat recaptures only the bounded $N = 2$ GDN cohort
Prefix immutability	complete	pass	physical document-KV digests and GDN base guards across all three phases	one frozen configuration/geometry
Private ownership	complete	pass	all-pairs request–base and request–peer range replay at the primary registered transition; actual-builder, serializer, and direct-clone receipts for the selected cell	pointer-free conservative intervals; stronger live binding covers only six selected rows per phase in one cell
Tail-safe append	complete	pass	one-time 127-token private-tail copy before append	one partial-tail geometry
Host-side launch/config.	complete	pass	per-call ID, selected kernel-cache artifact/configuration, autotuner outcome, GPU/stream, and post-return seal; source-bound GDN eager route	normal launcher return, not device completion; no driver/device, compiled-GDN, or ATen/CUDA attestation
Cross-arm equivalence	complete	pass	tokens, shape/dtype-bound CPU-FP32 logit digests, logical KV, and GDN	one primary sequential schedule
Cross-N consistency	complete	pass	288 all-arm adjacent-fan-out comparisons over 72 rank–query identities	relational, not an independent oracle

Selected numerical oracles, constructed controls, lifecycle/scheduler checks, the independent census and dual-producer repeat, bounded two-stream and Falcon cohorts, and the historical reproduction are supporting checks rather than additional contract targets. Their captured-input, same-adapter, process-separated-but-shared-IPC, second-configuration, and schedule boundaries are reported separately.

E POINTER-FREE STORAGE WITNESS SCHEMA

The pass predicate. For case specification Σ, execution trace τ, mandatory slots $M_i(\Sigma)$, and replay predicates Φ_i,

$$\mathrm{Pass}_i(\tau) \iff [\mathrm{Coverage}_i(\tau) = \mathrm{complete}] \wedge \mathrm{Bind}_\Sigma(\tau) \wedge \bigwedge_{\phi\in\Phi_i} \phi(\tau), \tag{4}$$

where completeness quantifies over $M_i(\Sigma)$ — every mandatory slot present, unique, unmodified — and Bind_Σ requires each slot to resolve to the live tensor its receipt names. Coverage is therefore separate from verdict: a missing, duplicated, modified, or unbound mandatory receipt cannot silently pass. The workflow freezes the run, captures setup/transition/final digests, ranges, calls, semantics and allocator snapshots, then replays every applicable predicate (Table 26).

Target-to-record dependencies. The following matrix is normative. I, L, E, and F denote the Identity, Lifecycle, Execution, and Target/fault classes in Table 26; parenthesized entries name mandatory fields. Missing any marked field makes that target open. A preregistered geometry may make tail safety not applicable. Target 5 additionally requires an exact per-call ID, selected kernel-cache artifact/configuration, autotuner selection or explicit no-autotuner record, assigned GPU/stream, and post-return seal; missing any applicable field leaves target 5 open. A missing F field invalidates the corresponding fault row, but does not change the separately evaluated seven-target witness.

Enumerated evidence units and call counts. Section 5.5 states the verdicts; the counts behind them are collected here, each tagged with its execution. *Dispatch rerun.* Host-side receipts for 209,920 attention calls record the selected hashed Triton artifact/configuration and the autotune/no-autotuner decision immediately before the original Python launcher, then seal after normal return; another 635,520 receipts record the mutually exclusive GDN eager route and functional cache rebind. The totals factor as $20{,}992 \times 10$ attention calls and $20{,}992 \times 30$ request-GDN calls plus 192×30 one-time prefill calls. They establish host-side selected launch/configuration provenance only, not device-side operator identity, execution, or completion, compiled GDN, underlying ATen/CUDA, or driver/device binaries. *Live-binding rerun.* Table 7 defines each evidence unit. The totals are $8 \times 3 \times 6 = 144$ selected-owner-row checks, $8 \times 3 \times 540 = 12{,}960$ full serializer/live-object rows, $8 \times 480 = 3{,}840$ direct clone edges, and $8 \times 3 = 24$ stable phase artifacts, where the factor 3 is the setup, post-transition, and post-generation *phases* — not the three resident counts that appear in the $8 \times 3 \times 4 = 96$ configuration product of Section 5.5. Here $60 = 30 \times 2$ persistent layer/family coordinates (thirty GDN layers $\times$ recurrent and convolution families) give the $480 = 8 \times 60$ direct clone destinations, and each phase contains the full $540 = (1 + 8) \times 60$ serializer-row universe, one persistent plus eight request owners, with six owner-row anchors. All 96 separately enumerated clean-memory calls occur outside the active selected context; zero selected-context overlaps means only that this context never entered an allocator cell, not that the wrapper was absent or perturbation-free. *Primary rerun.* The 288 adjacent-fan-out comparisons of Section 5.5 span 72 rank–query identities across all arms, and the borrowed-GDN arm's own allocator endpoints are $4.890\,\mathrm{GiB}$ final and $2.229\,\mathrm{GiB}$ shared-document (Table 19). The primary factorial's preregistered candidate-import-free NumPy oracle checks four GDN transitions at global layer indices $0, 10, 20, 38$ against four wrong recurrences (Table 30), and its supporting cohorts derive 1,080 GDN observations and 96,660 relations from a source-distinct expected-slot census (Appendix H).

Target	I	L	E	F	Blind predicate
Frozen identity	code, model, data, protocol; role map	—	—	N/A	exact bytes and complete roles
Prefix immutability	document/base roles; start digests	setup and final guards	canonical content digests	N/A	start = final
Private ownership	request/base/private roles	setup, registered-transition, final bindings	normalized storage ranges	N/A	non-vacuous all-pairs disjointness
Tail-safe append	page geometry; document/private roles	first append event	ordered append and copied-tail receipt	N/A	one private copy before write
Host-side launch/config.	callable, source, protocol, selected artifact/config.	post-return seal; GPU/stream assignment	ordered call, query, mask, position, append, scale, autotuner outcome, eager route	N/A	exact per-call binding; no fallback or route contamination
Cross-arm equivalence	factor, query, and schedule identities	final bindings	token, logit, logical-KV, GDN records	N/A	canonical semantic equality
Cross-N consistency	query and fan-out identities	—	the same canonical semantic records	N/A	smallest-N equals each larger N
Fault sensitivity (supporting)	frozen target/gate map	matched-control rebuild and cleanup	injected outcome and failed predicates	required	target, injection, expected gate, outcome

Each tensor-view row has the exact fields `storage_id`, `shape`, `stride`, `storage_offset`, `dtype`, `device`, `storage_nbytes`, `tensor_nbytes`, and `content_sha256`; absolute addresses are forbidden. Within each phase, the producer assigns `storage-0000`, `storage-0001`, and so on in first-observation order to backing storages. Reusing an ID with a different device or storage size is invalid. Phase capture IDs are unique, whereas request and persistent lifecycle-guard IDs must remain stable across setup, the registered transition, and final generation.

Let element size be e, storage offset o, shape s_i, and stride t_i. The independent replay initializes $m = M = o$ and, for every dimension, computes

$$d_i = (s_i - 1)t_i, \qquad m \mathrel{+}= \min(d_i, 0), \qquad M \mathrel{+}= \max(d_i, 0). \tag{5}$$

The conservative half-open byte interval is

$$I = [em,\, e(M + 1)), \tag{6}$$

which must be non-negative and contained in `storage_nbytes`; also `tensor_nbytes`$= e \prod_i s_i$. Two rows overlap iff their normalized storage IDs match and $I_a^{\text{start}} < I_b^{\text{end}}$ and $I_b^{\text{start}} < I_a^{\text{end}}$. Exact aliasing additionally requires equality of interval, geometry, dtype/device, storage and tensor sizes, and content digest. For correctly live-bound rows at one capture, if represented views sharing physical backing receive the same normalized ID and their shape, stride, offset, and element size exactly describe affine-strided views, I contains every touched byte. The overlap test then cannot miss a shared byte between represented rows, although interleaving may cause false positives. Omitted rows, between-capture changes, and wrapper or storage-ID violations remain outside the claim.

At setup, every recurrent-state coordinate must either exactly alias the persistent state (borrowed-base arm) or be disjoint from it (materialized arm). At the primary registered 32-token transition boundary, a completed request's 60 mutable rows must be disjoint from the base and every peer; incomplete borrowed requests must remain exact aliases. At final, all request-local GDN states must be disjoint from the persistent GDN base and every peer; document KV remains intentionally shared. Independent binding-token receipts additionally require unchanged tokens for incomplete requests and changed binding/storage tokens for completed requests. The package applies this schema to all 96 timelines and 288 phase artifacts; nine adversarial tests cover exact aliasing, partial overlap, adjacent intervals, negative strides, conflicting ID reuse, pointer injection, capture/guard reuse, and cross-coordinate request–base and request–peer aliasing.

F Executed Fault Rows and Invariant Map

The row-level campaign is kept out of the main paper because it supports diagnosis and audit coverage rather than a separate headline result. Separately, the CPU governance suite passes 17 tests and rejects 28 explicit serialized-artifact tamper cases plus its registered helper/launcher faults. That result concerns schema and orchestration fail-closure only; it is not GPU semantic or performance evidence.

Table 11: Separate primary all-gates-on reference. Every separately rebuilt matched-clean cell passed; every mutant reached its target gate fixed before execution and was restored before disposal.

ID	Injected live fault	Expected/observed gate	Primary outcome
M1	reservation alias	`KV_RESERVATION_DISJOINT`	target gate reached
M2	sequence swap	`KV_SEQUENCE_ID`	target gate reached
M3	omit tail COW	`KV_TAIL_COW`	target gate reached
M4	GDN–base alias	`gdn_completed_vs_base_disjoint`	target gate reached
M5	GDN peer alias	`gdn_completed_vs_peers_disjoint`	target gate reached
M6	position off-by-one	`POSITION_CANONICAL_VALUES`	target gate reached
M7	materialized mask	`MASK_CONTRACT`	target gate reached
M8	wrong callable	`KERNEL_CALLABLE_ID`	target gate reached
M9	dense KV view	`KV_PAGED_VIEW`	target gate reached

Table 12: Same-system outcomes for nine designed faults. The primary campaign enables all gates; a separate target-gate-suppression campaign suppresses only the named gate. "Same" is exact and "N/O" an earlier stop. Rows are not a detection rate; M8's injected sentinel is not a production detector.

Fault / target contract	All gates on	With target gate suppressed	Token / FP32 logits
M1 Reservation ownership	target gate	completes	Same / Same
M2 Sequence binding	target gate	completes	Same / Same
M3 Tail COW	target gate	other audit: active-block ownership	N/O / N/O
M4 GDN base isolation	target gate	other audit: peer isolation	N/O / N/O
M5 GDN peer isolation	target gate	completes; logits change ($L_\infty = 0.75$, rel. $L_2 = 0.0314$)	Same / Changed
M6 Positions	target gate	completes	Same / Same
M7 Mask contract	target gate	completes	Same / Same
M8 Callable identity	target gate	injected payload sentinel	N/O / N/O
M9 Paged-KV view	target gate	paired-view production assertion	N/O / N/O

Designer–executor-separated fault detail. The five fixed per-fault outcomes summarized in the main text are shown below; they remain constructed controls rather than a defect-population sample.

Table 13: Designer–executor-separated constructed fault outcomes. A separate designer saw only a frozen earlier manuscript PDF and fixed all five definitions, loci, payloads, clean requirements, and expected first predicates before executor preparation or candidate output. The cases are separated from executor preparation and outcome inspection, not from the predicate vocabulary. All matched clean cases pass; every mutant completes its registered horizon and is rejected first by the frozen primary predicate. Semantic comparisons are secondary: "Same" means canonical exact equality with the matched clean case, KV/GDN denote terminal logical state, and "N/C" means full-logit call cardinality is not comparable. These five fixed outcomes are not a population detection rate.

Frozen fault	Isolated mutation	First failed predicate	Tokens / logits / KV / GDN
HF01 delayed tail detach	append write precedes the otherwise present tail copy	tail-copy-before-first-write	Same / Same / Same / Same
HF02 inactive-lane scribble	one bit in the unused lane of a shared physical K page	physical-prefix immutability	Same / Same / Same / Same
HF03 duplicate commit	one correct feedback call commits twice; first output discarded	ordered-call cardinality	Same / N/C / Changed / Changed
HF04 effective-scale drift	first registered attention scale is exactly doubled	attention effective scale	Changed / Changed / Changed / Changed
HF05 stale GDN binding token	storage correctly rebinds; lifecycle token remains stale	completed binding-token advance	Same / Same / Same / Same

Table 14: Representative failure families and intended checks. Rows naming M1–M9 refer to the nine executed primary-factorial live mutants; the other rows summarize schema, relational, numerical, or replay checks. None is a complete oracle.

Check family	Intended sensitivity and observed boundary
Frozen identity	wrong model/data/query bank, changed protocol, stale shard (schema adversaries)
Prefix immutability	M3 omits partial-tail COW; physical document digest and ownership gates fail
Private ownership	M1 reservation alias, M4 request–base GDN alias, M5 request–peer GDN alias
Sequence/view binding	M2 sequence swap and M9 dense-KV replacement
Host-side selected launch/configuration	M6 position offset, M7 materialized mask, M8 wrong callable
Factorial exactness	behavior changes when either KV or GDN setup ownership changes
Bounded numerical oracle	numerical error in a selected fused full-attention result
Raw replay	internally inconsistent trajectory, storage, allocator, or aggregate fields

The primary all-gates-on execution is a first-gate localization study. Every registered gate fires before the injected path produces a post-injection token, logit, logical-KV digest, or GDN digest; its output cells are therefore N/O rather than misses. Lifecycle/ownership/storage receipts supply the first gate for M1 and M3–M5; request-binding/call receipts supply it for M2 and M6–M9. The full audit rejects all nine at their predeclared gates under passing matched controls. The separate target-gate-suppression matrix then suppresses one named gate per mutant while leaving the remaining observers active. It provides the per-fault comparison in Table 12, but does not estimate a population rate, unseen-fault coverage, or false-positive rate.

G Relation to Prior Work and Audit Obligations

This appendix carries the attributions and distinctions Section 2 points to.

The established quantization surfaces. These methods pack one state family — attention keys and values — produced on a single execution path, whereas a hybrid split-replay entry mixes a boundary residual, lower-layer attention KV, and mutable convolution and recurrent state, and we report Store for the whole entry (Section 4.1) rather than the residual alone, which CoMem retains unquantized in BF16 (Liu et al., 2026a). KIVI, KVQuant and XQuant fix the grouping axis or the tensor quantized (Liu et al., 2024; Hooper et al., 2024; Tomar et al., 2025); KVTuner and AsymKV assign layer-wise mixed-precision KV bit pairs offline (Li et al., 2025a; Tao et al., 2025; Yang et al., 2024); Palu, SqueezeAttention and HeadKV allocate rank or budget rather than bits (Chang et al., 2025; Wang et al., 2025; Fu et al., 2025); and Quamba, Quamba2 and MambaQuant quantize selective-SSM activations and per-state-group recurrent inputs, including for a hybrid backbone (Chiang et al., 2025b;a; Yue et al., 2025). Our bit vector $\mathbf{b}$ is one setting on this surface, not a new selection method.

The two differences from XQuant. The cut point is a depth-j boundary residual, not a layer input (Tomar et al., 2025), so what is recomputed is the whole suffix $[j, L)$ for a later query over a persisted document, not one layer's projections inside its own decode step. And the retained object is not one family: below the split a hybrid backbone still needs full-attention KV with convolution and recurrent state its kernels mutate in place, so the entry is heterogeneous and raises the ownership question of Section 4.3 that a dense activation cache does not.

Evaluation-methodology precedents. SCBench emphasizes cache lifecycles (Li et al., 2025b); metamorphic testing, MT-DLComp, and CRADLE motivate preserved relations and independent checks (Chen et al., 2018; Xiao et al., 2022; Pham et al., 2019); DNNMem separates memory denominators (Gao et al., 2020).

Table 15: Primary-source positioning. “Evidence focus” describes what the cited work evaluates; it is not a claim that the work lacks all other checks.

Work	Primary mechanism	Evidence focus / relation to this work
PagedAttention	Paged KV allocation and sharing (Kwon et al., 2023)	Serving throughput and memory utilization; supplies the block-sharing substrate.
SGLang / Prompt Cache	Prefix reuse in structured or modular prompts (Zheng et al., 2024; Gim et al., 2024)	Runtime performance and output behavior; establish reuse and positional handling as prior art.
CoMem	One persistent residual per token at split j; bounded selection; suffix replay (Liu et al., 2026a;b)	Supplies the split-replay foundation; we do not claim that primitive as new.
XQuant	Quantized layer-input activation cache with key/value rematerialization at decode (Tomar et al., 2025)	Closest prior art on the quantize-an-intermediate-and-recompute idea, which we therefore do not claim; it cuts inside a layer on a dense backbone, whereas we cut at a depth-j boundary and retain heterogeneous lower attention KV plus mutable convolution/recurrent state. No head-to-head comparison is reported.
Q-COMEM (ours)	Joint low-bit split residual and required lower KV/convolution/recurrent state	Adds complete hybrid-state packing, a capacity–latency–quality evaluation, and an auditable request-local ownership boundary; not a model-weight compressor or production scheduler.
Marconi	Admission and eviction for hybrid-model prefixes (Pan et al., 2025)	Hybrid cache hit rate and time to first token; closest state-heterogeneous caching system, but a different policy-level contribution.
SCBench	Lifecycle-oriented KV-cache benchmark (Li et al., 2025b)	Breadth over tasks and cache phases; exposes breadth absent from our bounded declared configurations.
MT-DLComp / CRADLE	Metamorphic compiler testing and cross-backend differential checking (Xiao et al., 2022; Pham et al., 2019)	Direct testing precedents; our audit adds ownership witnesses and a layerwise dense oracle, but not an independent end-to-end backend.
DNNMem	Component-wise GPU-memory accounting (Gao et al., 2020)	Supports denominator separation; it is an estimator, not a request-ownership audit.
FORKAUDIT	Typed, phase-indexed ownership/call/accounting trace with explicit coverage semantics	Validates the ownership component through a mutation-tested KV-by-GDN factorial, bounded host-side call receipts, a preproducer expected-slot census, numerical sweeps, and one bounded Falcon-H1 comparison; it is not a security attestation or general end-to-end correctness oracle.

Table 16: Closest established precedent for each audit obligation. The final column states the narrower integration contributed by FORKAUDIT; it does not claim that any individual mechanism is new.

Audit obligation	Closest established precedent	Integrated obligation in this work
Frozen identity	Controlled variants in metamorphic and differential testing (Chen et al., 2018; Xiao et al., 2022; Pham et al., 2019)	Bind model, data, query bank, callable contract, and protocol bytes to every ownership receipt.
Document immutability and tail COW	PagedAttention block sharing and copy-on-write for shared prefixes (Kwon et al., 2023)	Replay physical document pages, padding, active tables, and the partial-tail detach event.
Recurrent-state ownership	Hybrid recurrent states and their in-place update constraint (Yang et al., 2025; Pan et al., 2025)	Witness request–base and request–peer ranges before and after the frozen schedule's registered transition boundary.
Host-side selected launch/configuration	Cross-backend differential checking exposes implementation-dependent behavior (Pham et al., 2019)	Hold the backend fixed; record callable, positions, mask, scale, and append history; and bind the selected Triton kernel-cache artifact/configuration per attention call while keeping GDN at an explicit eager-route boundary.
Cross-arm and cross-N relations	Metamorphic relations over semantics-preserving transformations (Chen et al., 2018; Xiao et al., 2022)	Treat ownership policy and resident fan-out as transformations that must preserve registered request observables.
Bounded numerical oracle	Independent backend comparison (Pham et al., 2019)	Reconstruct selected attention inputs for dense IEEE-FP32 math and selected post-normalization GDN inputs for a candidate-import-free NumPy recurrence.
Memory denominators	Component-wise GPU-memory accounting (Gao et al., 2020)	Keep page geometry, allocator current/peak, and unmeasured process/service capacity as separate claim domains.

H SUPPORTING CONTROL EXPERIMENTS

Mac common-mode motivation. The archival Apple-M4-Pro control is kept unpooled from the H20 experiments and serves only to show why split-family agreement does not replace a dense or independently implemented oracle.

Table 17: Unpooled Mac motivation context (Apple M4 Pro, five fresh MLX processes; median over 30 query executions at split depth 7). Vanilla dense is the external reference. Q-COMEM BF16/Q8/Q4 variants use the same selected documents. Store size is the six-document residual corpus; online time includes query Write, dequantization, and suffix Read. This small teacher-forced demonstration set is not primary ForkAudit evidence and supports no cross-hardware or production-speed claim.

Method	Store (MiB)	Online (ms)	Speedup	Top-1 vs. dense
Vanilla dense prompt	–	194.70	1.00×	100.00%
Q-COMEM BF16 split	1.998	168.88	1.15×	71.89%
Q-COMEM Q8 split	1.061	169.17	1.15×	71.89%
Q-COMEM Q4 split	0.562	169.25	1.15×	71.89%

Q8/Q4 retain 100% top-1 agreement with the BF16 split path, yet all split rows inherit only 71.89% agreement with vanilla dense. The speed column is a small teacher-forced mechanism benchmark, not production RAG throughput.

Historical alias regression and repair. The pre-fix, repaired, and materialized lanes isolate one previously observed recurrent-state alias. This is a fixed mechanism study, not a natural-bug frequency or detector-recall estimate.

Table 18: Retrospective reproduction and repair of one historical alias regression. Ranks 0–2 reuse three archived input coordinates; ranks 3–7 are additional frozen inputs, not additional defects or statistical replicates. Semantic columns compare each path with a fresh materialized control. "Base inv." is within-path persistent-base immutability, retained as a conventional state-invariant catch.

Coordinates	Path	Ownership result	Token	FP32 logits	Request GDN	Logical KV	Base inv.
Archived (3)	Pre-fix	binding gate 3/3	exact 3/3	exact 3/3	exact 3/3	exact 3/3	fail 3/3
Archived (3)	Repaired	storage-clean 3/3	exact 3/3	exact 3/3	exact 3/3	exact 3/3	pass 3/3
Additional (5)	Pre-fix	binding gate 5/5	exact 5/5	exact 5/5	exact 5/5	exact 5/5	fail 5/5
Additional (5)	Repaired	storage-clean 5/5	exact 5/5	exact 5/5	exact 5/5	exact 5/5	pass 5/5

Primary allocator endpoints and fixed-case detection detail. Token equality catches 0 of 5 semantics-completing mutants and exact logits 1 of 5 (Table 12); in the no-injection defect, output and terminal request state stay exact in 8/8 cells under persistent-base corruption that Table 18's invariant also catches on all eight pre-fix cells. The four ownership cells below use one frozen post-priming allocator denominator and are kept separate from deployment Store. Against the appropriate reference the ownership component saves nothing: Table 19's paged-prefix baseline is the closest same-stack system, and at $N = 32$ our audited fork ties it exactly ($2.229\,\mathrm{GiB}$ final, $2.843\,\mathrm{GiB}$ peak) while being 1.950 against $0.019\,\mathrm{GiB}$ worse on generation increment, so the design buys auditable request-local state, not fewer allocator bytes. The 54.5%/42.2% figures are measured against a different arm: with materialized GDN setup, shared-document rather than full-copy KV cuts the final allocated delta from 4.901 to $2.229\,\mathrm{GiB}$ (54.5%) and the peak from 4.920 to $2.843\,\mathrm{GiB}$ (42.2%) above the frozen post-priming baseline. That full-copy arm is a controlled ownership factor, not an optimized baseline: no serving engine copies a shared prefix per request, and the saving is block sharing (Kwon et al., 2023).

Table 19: Primary 8×H20-3e allocator results at $N = 32$ (median across ranks; rank values coincide). Full-copy is the factorial control; the paged-prefix row is the closest same-stack prefix-sharing baseline. The remaining rows isolate recurrent-base ownership. All cells retain the persistent document source. Values are GiB above the frozen post-priming baseline; generation increment is the additional peak above allocation present at generation start.

Empirical arm	KV / GDN setup	Final	Peak	Gen. incr.
Full-copy factorial control	Full-copy KV / Materialized GDN base	4.901	4.920	0.019
GDN ownership ablation	Full-copy KV / Borrowed GDN base	4.890	4.907	1.951
Paged-prefix baseline	Shared-doc KV / Materialized GDN base	2.229	2.843	0.019
Audited hybrid fork	Shared-doc KV / Borrowed GDN base	2.229	2.843	1.950

Published systems context. Table 20 reports each cited system only in its native protocol and keeps all non-commensurate metrics outside our matched panels. This unpooled table is literature context, not ForkAudit evidence.

Table 20: Unpooled published-systems context in each paper's native protocol. These values show the scale and evaluation breadth of related systems, *not* a cross-paper leaderboard or ForkAudit evidence: models, hardware, batching, baselines, and metrics differ. Consequently, no row is merged with our same-stack H20 Table 23, and the absence of a common score must not be read as a negative result.

System	Native method and study setting	Reported efficiency in its native protocol	Quality and comparability boundary
PagedAttention / vLLM (Kwon et al., 2023)	Paged KV allocation plus serving engine *Setting:* LLaMA-7B on A10G and LLaMA-13B on A100-40GB; vLLM serving engine	2–4× serving throughput at similar latency vs. FasterTransformer/Orca	No common LongBench F1; exact cache-management method
Prompt Cache (Gim et al., 2024)	Modular attention-state reuse *Setting:* Llama2/MPT/Falcon; RTX 4090, A40, and A100; HuggingFace-style prototype	1.5–3.1× TTFT (CPU-resident modules); 3.7–11.7× (GPU-resident) on its 12-dataset LongBench suite	LongBench accuracy evaluated, but with different models and hardware
SGLang / RadixAttention (Zheng et al., 2024)	Prefix reuse plus structured-program runtime *Setting:* Llama-7B/Mixtral-8x7B and structured programs; A10G/A100; SGLang	Up to 6.4× throughput vs. prior inference systems	Application-suite result; no common LongBench F1
ChunkAttention (Ye et al., 2024)	Prefix-aware KV chunks and attention kernel *Setting:* attention microbenchmarks; mainly A100-80GB, plus A100-40GB/RTX 4090; CUDA 11.8	3.2–4.8× attention-kernel speedup for 1K–4K shared prompts	Kernel result; no end-to-end common task score
Hydragen (Juravsky et al., 2024)	Exact batched shared-prefix attention *Setting:* CodeLlama-13B/34B shared-prefix batches; A100 (with H100/L40S microbenchmarks); custom HuggingFace runtime	Up to 32× CodeLlama-13B end-to-end throughput in large shared-prefix batches	Exact attention construction; different model and batching regime
Palu (Chang et al., 2025)	Low-rank latent KV with reconstruction/fusion *Setting:* Llama/Mistral/LongChat quality studies; Llama-2-7B latency on one RTX 4090; HuggingFace plus custom kernels	50% KV compression and up to 1.89× RoPE-attention speedup; up to 2.91× with quantization	Llama/Mistral/LongChat native study; our bounded Qwen3.5 operator transfer is reported separately
Preble (Srivatsa et al., 2025)	Distributed prompt-sharing scheduling *Setting:* Mistral-7B/Llama-3-70B request traces; two–eight GPUs across A6000/H100 clusters; distributed serving	1.5–14.5× average-latency and 2–10× p99-latency improvement	Distributed serving traces; no common task score
Marconi (Pan et al., 2025)	Admission/eviction for hybrid-model prefixes *Setting:* Attention–Mamba2 7B geometry; LMSys/ShareGPT/SWE-bench traces; CloudLab policy simulation	Up to 34.4× token-hit rate and up to 71.1% or 617 ms lower TTFT	Hybrid models and trace-driven policy study; no common LongBench F1
KVTuner (Li et al., 2025a)	Offline sensitivity-aware search over layer-wise mixed-precision KV bit pairs *Setting:* Llama-3.1-8B-Instruct and Qwen2.5-7B-Instruct; mathematical- and general-reasoning suites	Nearly lossless 3.25-bit mixed-precision KV for Llama-3.1-8B-Instruct and 4.0-bit for Qwen2.5-7B-Instruct; up to 21.25% higher maximum inference throughput vs. KIVI-KV8	Layer-wise bit search for a dense KV cache; it does not quantize a split residual or recurrent state, and no common Qwen3.5 slice exists
HYPIC (Liu et al., 2026c)	Position-independent caching for hybrid full/linear-attention models *Setting:* Qwen3.5-35B-A3B main panels at TP=2 on 8 H20-3e; SGLang 0.5.14	3.25× lower TTFT on average and 1.66× higher QPS over Prefix Cache	Four models/five workloads; our same-slice TP=1 timing and retained-state controls are reported separately

Same-checkpoint transfer detail. Table 21 is the matched complement to the published context table: it uses our frozen Qwen3.5 checkpoint and H20-3e, binds the official Hydragen and Palu source revisions, and independently replays all raw operator sidecars. Because it does not execute all layers or LongBench, it is kept separate from Table 23.

Table 22: The latency evidence, with the dispersion earlier versions of this panel averaged away. (a) The eight-item Qasper/2WikiMQA execution (source indices 6–9 of each dataset), every arm capped at 32 generated tokens, never pooled with the 60-item panels (Section 5.1). (b) The per-repeat spread of every timing cell of the operating-point run. Both use Qwen3.5-35B-A3B revision `59d61f3`, PyTorch 2.11.0+cu129, Transformers 5.14.1, CUDA 12.9, one H20-3e per item, a 4,096-token cap, greedy decoding and three timing repeats.

(a) Eight-item execution; every cell a median over the eight documents, three repeats each

Retained state	Store (MiB/doc)	Reduction	TTFT (s)	TPOT (ms)	tok/s†	F1 (0–100)
No-cache dense recompute	0.00	–	0.649	648.75	1.36	39.42
Full-prefix, native dtype	140.34	reference	0.163	54.11	13.38	39.14
Q-CoMem split Q8	15.89	88.68%	0.673	55.44	7.15	39.14
Q-CoMem frozen Q4/Q4/Q8	10.01	92.87%	0.674	54.97	7.19	39.01
Q-CoMem calibrated per-layer	9.74	93.06%	0.673	55.07	7.19	39.11
Q-CoMem split Q4	8.41	94.01%	0.674	55.02	7.19	36.09

(b) Operating-point run, 60 items × 3 repeats, seed 20260903; min / median / max over the three repeats

	TTFT (s)			TPOT (ms)			Median generated
Arm	min	med	max	min	med	max	tokens (cap 32)
No-cache dense recompute	0.2587	0.6482	1.4729	50.60	53.46	86.63	4
Full-prefix, native dtype	0.1500	0.1610	1.0891	50.53	53.45	404.01	4
Full-prefix Q8 (uniform 8-bit)	0.1545	0.1646	1.3874	50.73	53.46	89.93	5
Q-CoMem frozen Q4/Q4/Q8, $j = 13$	0.3192	0.5826	1.5740	51.51	54.18	89.34	4
Q-CoMem frozen Q4/Q4/Q8, $j = 7$	0.3527	0.6683	1.5296	51.56	54.19	86.84	4

Aggregation. Block (a) reports medians over the eight documents and no dispersion, none having been recomputed for that execution; block (b) the minimum, median and maximum of the three per-repeat values. Table 4's operating-point timing cells are that run's recorded aggregates over the same 900 rows under a rule it did not record, so they are not the block-(b) medians (full-prefix reads 0.181 s there against 0.161 s here), though each falls inside the matching min–max range. †Throughput is recorded, not derived: no single generated-token count n satisfies $n/(\mathrm{TTFT} + (n-1)\,\mathrm{TPOT})$ across block (a) ($n \approx 5.3$ full-prefix, 7.3–7.4 Q-CoMem, none at or above one token for dense, whose own timings give 1.541 tok/s at the cap against 1.36 printed), and in the operating-point run the reconstruction from matched per-repeat values still misses recorded throughput by a median 7.6–18.8% per arm, so the column must not be inverted for a token count. Store is the deduplicated sum of the tensor-storage bytes one retained entry owns (Section 5.1); block (a)'s reference is the entry in native dtypes and every "split" arm retains a $j = 7$ entry (Appendix C); its calibrated per-layer row is smaller and better than its frozen row, the opposite order to the 60-item panel (Section 5.4). Recall was measured in no cohort.

Table 21: Bounded same-checkpoint H20 transfers. Hydragen uses frozen Qwen3.5 layer-3 post-RoPE attention; Palu uses eight 256-token PG19 calibration rows and one disjoint held-out row. Errors are relative ℓ_2 against the matched uncompressed operator; state ratios are logical selected-operator ratios, not process memory or end-to-end results.

Port	Setting	Matched numerical result	Logical state result	Timing / boundary
Hydragen port	$N = 8$	Hydragen–dense rel. ℓ_2 = 0.00255	8.50 vs. 64.48 MiB (7.59× smaller)	0.287 vs. 0.0965 ms; no speedup
Hydragen port	$N = 32$	Hydragen–dense rel. ℓ_2 = 0.00262	10.00 vs. 257.94 MiB (25.80× smaller)	0.290 vs. 0.259 ms; no speedup
Palu port	rank 64/head	held-out K/V $0.487/0.549$ (plain $0.711/0.839$)	512 vs. 2,048 B/token (4.00× smaller)	projection only; no fused kernel
Palu port	rank 128/head	held-out K/V $0.317/0.364$ (plain $0.543/0.668$)	1,024 vs. 2,048 B/token (2.00× smaller)	projection only; no fused kernel
Palu port	rank 192/head	held-out K/V $0.149/0.194$ (plain $0.341/0.459$)	1,536 vs. 2,048 B/token (1.33× smaller)	projection only; no fused kernel

Hydragen avoids prefix replication but has no speedup here. Palu improves over plain SVD, but residual error forbids an end-to-end quality claim. Neither is a jointly optimized combined system.

H.1 Extended Deployment and Serving Context

Table 22 is the latency panel of Section 5.3, printed here for space rather than in the body; the expanded table after it repeats panel (a) to place separately measured SGLang/HYPIC context beside it, and inherits panel (a)'s reporting caveats, including a throughput column that reconciles with no generated-token count. The blocks use different runtimes and harnesses, so timing and F1 are interpreted within block and are not pooled. These serving controls are not ForkAudit validation evidence and do not establish robust quality preservation.

Matched serving-framework and HYPIC context. Table 24 gives separately timed vLLM and SGLang cache-off/on controls plus the official-code HYPIC Full Recompute, Prefix Cache, and

Table 23: Illustrative same-checkpoint H20 retained Store and mean-F1 point estimates on an eight-item Qasper/2WikiMQA slice. All rows use Qwen3.5-35B-A3B, one H20-3e per item, a 4,096-token cap, greedy decoding, and at most 32 generated tokens. The Transformers (three repeats/item) and HYPIC (Liu et al., 2026c) blocks are unpooled; HYPIC timing/F1 and Store use separate 24- and 16-cell TP=1 cohorts. Timing is compared only within blocks; no robust-quality, capacity, or ForkAudit-ownership claim is made.

Method	Persistent state	Store (MiB)	TTFT (s)	TPOT (ms)	tok/s	F1
Transformers 5.14.1 in-process deployment cohort						
No-cache dense recompute	–	–	0.649	648.75	1.36	39.42
Full-prefix KV cache	Q16	140.34	0.163	54.11	13.38	39.14
Q-CoMem state	Q8	**15.89**	0.673	55.44	7.15	39.14
Q-CoMem mixed state	Q4/Q4/Q8	10.01	0.674	54.97	7.19	39.01
Q-CoMem per-layer mixed	mixed	**9.74**	0.673	55.07	7.19	39.11
Q-CoMem state	Q4	8.41	0.674	55.02	7.19	36.09
HYPIC commit 98147c0 timing/F1; receipt-instrumented Store; SGLang 0.5.14, TP=1						
Official HYPIC code: full recompute	–	–	0.217	49.21	14.77	39.14
Official HYPIC code: prefix cache	Radix prefix	139.53	0.072	50.15	19.07	39.14
HYPIC (transition + seam 8)	segment transition	324.09	0.101	52.13	17.67	39.63

Bold marks the two illustrative Store–F1 points; these are not a headline validation result. Relative to full-prefix Q16, Q8/per-layer mixed reduce Store $88.68\%/93.06\%$; unrounded mean-F1 deltas are $0.000/-0.022$ (39.115 vs. 39.137 for mixed). "–" means no retained entry; Q16 is the unpacked reference, BF16 for the residual and attention KV and the stack's native FP32 for GDN recurrent state. Full-prefix Q16 is Q-CoMem's matched semantic comparator; the archived rows show no-cache recompute sharing the same document/query boundary but emitting different greedy token sequences on a few items. Q4/Q4/Q8 gives residual/attention/linear bits. The per-layer policy (Q4 residual, layer bits $[8, 8, 4, 4, 8, 8, 8]$) was selected on a separate, disjoint four-item calibration set and frozen before this eight-item validation. Store is median retained-document tensor payload and excludes metadata, pools, process deltas, and capacity. Published HYPIC panels use TP=2; this adaptation uses TP=1.

full HYPIC arms under the shared Qwen3.5 streaming boundary. It remains a timing/quality-only panel and authorizes only within-framework interpretation; the separate 16-cell retained-state receipt cohort appears in Table 23. vLLM and SGLang each record 8/8 prefix hits and preserve 8/8 cache-off predictions. Within its TP=1 block, HYPIC gives 39.63, 0.101 s, and 17.67 tokens/s; published Qwen3.5 panels use TP=2, so this is an adaptation. Its 24 timing cells bind a 4,404-entry terminal artifact ledger. Across the 16 retained-state cells, Prefix Cache retains a median 139.53 MiB and HYPIC retains 324.09 MiB, against 15.89 MiB for Q-CoMem split Q8. The two sides do not use identical estimands: the SGLang-side receipts measure the byte-range union of the target entry's owned storages, whereas our printed column is the deduplicated sum of owned tensor-storage bytes (Section 5.1); on these eight documents the two coincide to the byte for every Q-CoMem configuration, which is what licenses placing the levels side by side, and not a shared definition. We report the three levels rather than their ratios: the SGLang payload is quantized to 128-token pages (across the eight documents it takes only three distinct values, all pairwise gaps integer multiples of the 1,310,720-byte ten-layer page) whereas the in-process payload is element-exact and varies continuously with document length. The one object both stacks hold — the retained full prefix for the same checkpoint, slice and cap — reads 139.53 MiB under the SGLang receipts and 140.34 MiB in-process, a 0.58% difference, which bounds the cross-runtime inclusion-rule discrepancy at that single design point. This remains single-stream; it does not measure continuous batching.

Table 24: Unpooled timing/quality-only serving context on Qwen3.5 and the same eight-item Qasper/2WikiMQA slice. Each row uses one H20-3e per item, a 4,096-token cap, greedy decoding, and at most 32 generated tokens. Times are OpenAI-compatible streaming client wall-clock; F1 is the eight-item mean. Hit is interpreted only against the disabled/full-recompute reference inside each framework block. No cross-framework speedup, retained-state, scheduler-throughput, capacity, or broad-quality claim is made.

Framework	Prefix reuse	F1	TTFT (s)	TPOT (ms)	tok/s	Hit
vLLM 0.26	off	39.63	1.445	56.31	4.70	–
vLLM 0.26	align on	39.63	0.179	53.99	14.00	8/8
SGLang 0.5.17	off	39.14	0.283	53.38	12.73	–
SGLang 0.5.17	Radix on	39.14	0.146	53.15	15.83	8/8
HYPIC code	full recompute	39.14	0.217	49.21	14.77	–
HYPIC code	prefix cache	39.14	0.072	50.15	19.07	8/8
HYPIC	transition+seam 8	39.63	0.101	52.13	17.67	8/8

vLLM uses experimental Mamba `align`; SGLang 0.5.17 uses `extra_buffer` Radix caching. The HYPIC rows use the authors' commit 98147c0 on SGLang 0.5.14, TP=1, full `transition_rope_recompute`, and an eight-token seam. Its full and prefix rows are controls from that same codebase; published TP=2 results and the separate retained-state receipt cohort are not inserted here.

Hybrid-prefix policy context. The separate preregistered Marconi trace extends the eight frozen workloads with a fixed synthetic follow-up turn and evaluates only prefix-policy token reuse under the artifact's native Attention–Mamba2 geometry. It neither reruns LongBench quality nor supplies Qwen3.5 serving speed, so it is not pooled with the preceding panel or the primary audit.

Table 25: Unpooled preregistered policy-simulator context on a 128-event synthetic multi-turn extension of the eight frozen workloads. Values are token-hit percentages under the official Marconi artifact's native Attention–Mamba2 geometry. The wrapper calls each policy's own eviction routine after insertion to close actual tree size to the declared budget. These are not Qwen3.5 quality, latency, or throughput measurements and do not validate the primary ownership audit.

Policy	5 GB	10 GB	20 GB
vLLM+	2.07	6.18	46.83
SGLang+	90.78	93.86	93.86
Marconi	**93.86**	93.86	93.86

Table 26: Normative record classes. "Mandatory" means that absence makes the corresponding target open rather than silently not applicable.

Record class	Mandatory payload	Independent replay
Identity	frozen code, model, data, protocol, and object-to-role map	exact bytes and role coverage
Lifecycle	setup, registered-transition, and final inventories and bindings	normalized ranges, aliases, and disjointness
Execution	ordered call/append receipts, semantic records, named allocator snapshots	frozen schedule, digests, and phase equations
Target/fault	predeclared target, gate, matched control, injection, and outcome	failed-predicate set and classified outcome

Aligned cancellation and reclamation lifecycle. The separate lifecycle-transfer cohort uses eight ranks, eight distinct PG-19 training books, an aligned 4,096-token prefix, and $N = 4$. It is not pooled with the primary factorial.

Table 27: Lifecycle-transfer obligations on the existing Qwen3.5/vLLM BF16-KV adapter. Counts are deterministic rank outcomes, not stochastic replicates.

Obligation	Result
Frozen static/code/model/weight/raw bindings	pass; 14/14 weights, 8/8 shards
Full-vocabulary logits and generated tokens vs. uninterrupted control	exact, 8/8
Final logical KV and final GDN state	exact, 8/8
Aligned prefix, immutable document, and zero partial-tail copy	pass, 8/8
Cancelled-slot zero-scrub and exact reservation reassignment	pass, 8/8
Stale epoch-0 rejection and matched epoch-1 acceptance	8/8; 0 escapes, 0 wrong gates

Scheduler-managed request-step interleaving. The separate formal extension keeps four requests resident, cancels slot 3 after its second round, zero-scrubs and increments its lease epoch, reassigns the exact reservation, and completes the replacement interleaved with the three survivors. It is not pooled with the primary factorial or the aligned lifecycle-transfer cohort.

Table 28: Formal scheduler-managed request-step interleaving on the frozen Qwen3.5/ForkAudit vLLM BF16-KV stack. Clean counts are deterministic rank outcomes; fault counts are preregistered expected-gate outcomes, not a detection rate.

Case	Frozen geometry or intervention	Outcome
Clean geometry 1	page 64; 3,985-token prefix; 17-token document tail; 18-event cancel/replacement lifecycle	all semantic/storage gates and uninterrupted-control comparisons pass, 8/8
Clean geometry 2	page 128; 4,033-token prefix; 65-token document tail; same lifecycle	all semantic/storage gates and uninterrupted-control comparisons pass, 8/8
FH1	dispatch cancelled epoch-0 lease	`STALE_SLOT_LEASE`, 16/16
FH2	dispatch lease under another request	`LEASE_REQUEST_MISMATCH`, 16/16
FH3	reclaim before zero scrub	`RECLAIM_NOT_ZERO`, 16/16

Candidate-import-free GDN transition oracle. The oracle uses one frozen PG-19 window and four query-time GDN transitions at global model-layer indices 0, 10, 20, and 38. A single preregistration frozen before the successful run binds the post-native-q/k-normalization boundary, scale $1/\sqrt{128}$, tolerances, and seeded faults. Its NumPy reference starts at that captured boundary; no candidate module is imported by the reference.

Table 29: PDF-visible oracle selection and replay boundary. Every listed row was locked before its successful candidate execution; neither oracle validates upstream activation construction or end-to-end model behavior.

Oracle	Frozen rows / selection rule	Captured boundary	Decision rule	Replay availability
Attention	rank r uses layer $4r+3$ and round r, for $r \in \{0,\dots,7\}$; request 0, $N=1$, shared-document KV + borrowed GDN; frozen rank/layer/round map	post-RoPE Q/K/V, positions, mask, scale, and ordered append shadows	dense CPU-FP32 rel. $L_2 \le 0.005$	8 raw rows and numerical sidecars; primary-package one-command replay
GDN	global model layers $0, 10, 20, 38$; four query tokens from one frozen PG-19 window	post-native-q/k-normalization recurrence; scale $1/\sqrt{128}$	output rel. $L_2 \le 0.005$; state rel. $L_2 \le 10^{-4}$; seeded fault must reject	4 clean + 4 fault decisions from 40 sidecars; separate one-command replay

Table 30: Bounded GDN recurrence oracle. Relative errors compare native outputs and final states with the candidate-import-free NumPy FP32 recurrence.

Global layer	Output rel. L_2	State rel. L_2	Clean	Seeded fault	Rejected
0	0.001584	1.23×10^{-7}	yes	omit decay	yes
10	0.001652	1.36×10^{-7}	yes	complement β	yes
20	0.001572	1.39×10^{-7}	yes	pre-decay memory	yes
38	0.001638	1.30×10^{-7}	yes	roll value heads	yes

Table 31: Preregistered expanded captured-boundary numerical sweep. Clean rows use candidate-import-free CPU/NumPy FP32 replay; every clean row has one seeded wrong-operator control. Attention begins after RoPE and GDN begins after native q/k normalization, so the table does not validate upstream activation construction, independent capture, or end-to-end model behavior.

Operator	Inputs	Layers	Clean rows	Positions / transitions	Max output rel. L_2	Controls rejected
Attention	2	10/10	20/20	160 positions	0.001897	20/20
GDN	2	12/30	24/24	192 transitions	0.002073 (2.08×10^{-7} state)	24/24

Full target-gate-suppression matrix. The preregistered target-gate-suppression execution contains all M1–M9 faults and a fresh matched clean path for each. Across eight distinct H20s, all 18 cases complete their declared horizon or an allowed classified stop, all 24 FP32 sidecars verify, and no case is operationally invalid. Five mutant paths complete semantics: token equality catches 0/5, exact logits catch M5 only ($L_\infty = 0.75$, relative $L_2 = 0.0314$), and M1/M2/M6/M7 leave both exact. M3/M4 reach other audit gates; M9 reaches the paired-view production assertion; M8 reaches the exact injected payload sentinel, which is not a production detector. The frozen aggregator initially read the inherited execution identifier from a detached-manifest field that the manifest schema omits. A disclosed postexecution wrapper derives that identifier from the canonical receipt embedded in all eight manifest-bound primary-package shards, changes only the generated comparison-row `run_id` from null to the preregistered value, and reproduces the summary byte-for-byte without modifying candidate outputs or frozen sources.

I Expanded Limitations and Claim Boundary

The primary factorial covers one model, one H20 hardware family, one page size, 16-bit KV, and $N \leq 32$. Its ownership conclusion is tied to one 4,095-token partial-tail geometry, a registered 32-token continuation block, and the subsequent frozen sequential schedule; it is not a claim about an arbitrary first token or transition length. The lifecycle cohort adds one aligned 4,096-token geometry and a fixed cancel/scrub/reclaim sequence. The scheduler-interleave extension adds two geometries and deterministic request-step interleaving, but those requests execute sequentially on one CUDA stream.

The separate two-stream cohort has two host workers and distinct CUDA streams with positive full-model-call interval overlap. Its intervals include queuing, resource waits, and host-enqueue gaps; there is no per-kernel overlap trace. Cancellation occurs only after synchronization, so the cohort does not test arbitrary or in-flight reclamation. The original out-of-process GDN observer removes same-process reconstruction and live judgment labels. The later preproducer census independently fixes the expected slot set, but correct slot-ID-to-live-tensor binding remains trusted; both sides also trust PyTorch tensor/storage and CUDA-IPC semantics, and mutation is paused for each snapshot. The separate selected-cell hook binds an actual builder return through the production serializer and direct clone lineage, but still runs with the producer in one honest process. These cohorts do not supply OS/driver allocation ground truth, independent receiver-side model execution, KV/kernel attestation, or protection against a transient write followed by restoration.

No cohort evaluates native continuous/ragged batching, eviction, multi-document serving, general scheduler cancellation, process-level NVML memory, OOM capacity, or a broad downstream benchmark. Full-copy KV and materialized GDN bases are controlled ownership factors rather than optimized production baselines. The GDN path is the Transformers Torch implementation, not an optimized recurrent kernel. The target-5 rerun closes only the declared honest-process host-side boundary: attention calls bind the selected Triton kernel-cache artifact/configuration and autotuner record, whereas GDN calls bind eager source routes. It does not attest device-side completion, compiled GDN or underlying ATen/CUDA operators, driver/device binaries, a malicious runtime, or another model/runtime/hardware stack. The live-binding rerun strengthens ownership provenance only in one selected cell and six rows per phase; other factorial cells and coordinates retain the trace-relative binding boundary.

The 60-item Qasper/2WikiMQA quality cohort is the broadest direct answer-quality evidence in this paper, but it is a single archival validation run on one model/H20 stack. Its raw shards and outcome calculations replay, whereas the original mutable code directory and absence of pre-run full source/model-weight ledgers prevent a source-frozen regeneration claim. Its paired interval does not establish equivalence or noninferiority. It records neither TTFT nor Recall. The separate systems execution uses the indices-6–9 subset of this cohort and supplies TTFT/TPOT/throughput point estimates, not an independent replication, population uncertainty, or a speedup over full-prefix reuse; its throughput column reconciles with no generated-token count, no per-repeat dispersion was recomputed for its cells, and it has no Recall measurement. The two panels are not pooled, and blank table cells preserve these gaps explicitly.

The auxiliary serving panel uses one request stream per engine and compares each cache-enabled framework only with its own disabled reference. It does not measure continuous batching, eviction, admission control, maximum concurrency, or best-tuned performance. CUDA graphs were disabled in both pinned runs; the reported values describe that fixed configuration only.

The HYPIC timing/quality comparison uses the authors' exact tracked commit on SGLang 0.5.14 in one fixed CUDA 12.9 / PyTorch 2.11 configuration, adapting Qwen3.5 execution from the paper's TP=2 to TP=1 so every arm uses one H20-3e per workload. Cache-hit authority comes from OpenAI response usage. Each of its 24 timing/quality cells has one formal measurement after a discarded, prefix-disjoint warmup. A separate receipt-instrumented run uses the same model/slice and fixed TP=1 serving configuration to produce 8 Prefix Cache and 8 HYPIC retained-state cells; each completed cell binds raw output plus pre-measurement and terminal ownership receipts, and all 16 pass an external hash-bound replay. The instrumentation adds target-entry byte-range receipts and a lifecycle-idempotency overlay; Store therefore does not come from the unmodified timing run. Neither cohort establishes scheduler throughput, QPS, capacity, the paper's TP=2 latency, or accuracy robustness beyond this eight-item slice.

The same-checkpoint related-work transfer is likewise operator-bounded: Hydragen uses one captured layer-3 shared-prefix attention family, and Palu uses one layer with eight calibration rows and one held-out row. Neither executes all model layers, a fused Palu kernel, LongBench, or a jointly optimized Hydragen/Palu-plus-CoMem serving path.

The page equations are deterministic identities, not learned laws; the primary factorial has one observation per rank–configuration cell and no stochastic replication. Most runtime tensors are represented by hashes and equality receipts rather than archived in full. The attention and GDN oracle rows are the exception and ship numerical sidecars, but cover only selected captured-input operator boundaries rather than an end-to-end model. The expanded numerical sweep covers two short inputs, all ten attention layers, and 12 of 30 GDN layers; it does not establish length/input robustness or full recurrent-layer coverage. Both the original and expanded GDN runs start after native q/k normalization and therefore do not validate that upstream boundary. Pointer-free normalized intervals plus raw-byte hashes make the primary captured storage evidence replayable. The preproducer census supplies the expected semantic slot set, and a second producer/observer execution reduces single-run capture risk, but both still trust correct slot-ID-to-live-tensor binding, IPC semantics, and paused capture; neither protects against a malicious producer. The selected-cell live binding directly checks the actual builder, serializer, and clone lineage, but shares the same honest runtime and does not generalize to unselected coordinates. The separate Falcon track compares against an official full-model reference only on its frozen Transformers/H20 naive path and supports no performance, memory, quality, or cross-runtime claim. The nine mutants are targeted positive controls; the target-suppression matrix adds a same-system comparison for those same designed faults, not blind or naturally occurring defects. Four complete with unchanged measured tokens and exact logits, one changes only full logits, and four stop earlier; N/O cells are not misses, while the injected M8 sentinel is not a production detector. This is not a general detection rate, a blind-fault result, or superiority over every conventional test suite. The scheduler extension's three frozen faults are expected-gate outcomes, not a detection rate or evidence of fault-set completeness. Broader systems claims would additionally require a native production serving scheduler, per-kernel overlap evidence, in-flight cancellation tests, optimized controls, alignment/length/input sweeps, and process-level capacity accounting.

The five designer–executor-separated controls were fixed before executor preparation, but remain deliberately constructed, fixed-stack mutations. Passing all five does not estimate unseen-fault recall, false-positive rate, or robustness to arbitrary runtime, model, schedule, or adversarial capture defects.

The retrospective case study concerns one organically encountered integration defect in our borrowed-state path. Only ranks 0–2 reproduce its archived input coordinates; ranks 3–7 are additional frozen inputs, and none is a statistically independent defect. The study does not estimate natural-defect prevalence or recall. Output and terminal-state differentials miss the alias, while a persistent-base content invariant catches it; FORKAUDIT contributes authenticated transition-time owner/layer/family localization rather than exclusive detection over all conventional testing.

J ARTIFACT AND CLAIM MAP

Artifact footprint. The source-complete primary package, derived summaries, raw ledgers, and expanded numerical sidecars are content-bound and locally replayable. The table below maps each manuscript claim to its machine-readable source and validation boundary; supporting measurements not used by a manuscript claim remain in the internal experiment registry.

Artifact aliases. Every package below is named by a stable neutral alias listed with its SHA-256 in the accompanying research package's `MANIFEST.json`. Aliases replace the internal directory names used during development; they carry no ordering, date, or platform information, and no path here resolves to an external service. A reviewer verifies each package by content hash rather than by path.

Table 32: Artifact and claim map. Relative paths are rooted at the accompanying research package. Completed rows authorize only their stated bounded claims. Metadata-redacted derivatives preserve numerical fields and state digests, while most full tensors still require the pinned environment for independent numerical recomputation.

Claim/evidence	Machine-readable source
Primary factorial replay package	`package/fa-01-primary-factorial/`
Replay and package manifest	`package/fa-01-primary-factorial/replay/run_replay.sh` and `package/fa-01-primary-factorial/MANIFEST.json`
Registered, derived, and raw evidence	`upstream/forkaudit-summary.json`, `derived/derived_summary_v2.json`, and `upstream/raw/` under the primary replay package
Executed source and storage tests	`executed_source/gpu/` and `replay/test_storage_witness.py` under the primary replay package
Identity, lifecycle, scheduler, and GDN extensions	`package/fa-08-terminal-closure/`, `evidence/lifecycle_transfer_reviewer_package/`, `evidence/round23_a2_scheduler_interleave/`, and `evidence/gdn_transition_oracle_preregistered_20260819d/`
Mac and H20 deployment controls	`package/qp-mac-motivation/`, `package/qp-08-timing/`, and their generators under `scripts/`
Q-CoMem 60-item Store–F1 validation	`package/qp-60-store-f1/`; exact raw-outcome replay: `replay/run_replay.sh`; numerical summary and claim boundary: `validation_report.json` and `claim_boundary.json`; registry entry `E-QCOMEM-60-VALIDATION-A`
HYPIC retained-state receipts	`evidence/experiment_registry.json`, entry `E-HYPIC-STORE-EXTERNAL-ACCEPTANCE`

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Table 32 continued.

Claim/evidence	Machine-readable source
Target-gate-suppression preregistration and raw ledger	`package/fa-02-gate-suppression/formal_run_20260824a/preregistration/detector-matrix-v2-plan.json` `package/fa-02-gate-suppression/formal_run_20260824a/receipts/raw-artifacts.sha256`
Target-gate-suppression summary and replay receipt	`package/fa-02-gate-suppression/formal_run_20260824a/detector-matrix-v2-summary.postexec-corrected.json` `package/fa-02-gate-suppression/formal_run_20260824a/postexecution-correction-replay-receipt.json`; disclosed field-source amendment: `package/fa-02-gate-suppression/postexecution-rr2-run-binding-correction.json`
vLLM, SGLang, HYPIC timing/quality, and Marconi controls	`package/rw-serving-controls/`; exact package members and one-command replays reside under this directory
Hydragen/Palu same-checkpoint transfers	`package/rw-operator-transfer/` and its `executed_source/` directory
Published context and table generator	`literature/reported_system_context.json` and `scripts/generate_related_work_context_table.py`
Out-of-process GDN observer	`package/fa-06-slot-census/formal_h20/result/preregistration/preregistration.json` `package/fa-06-slot-census/formal_h20/result/raw/out-of-process-gdn-capture.json` `package/fa-06-slot-census/formal_h20/result/replay/out-of-process-gdn-replay.json` `package/fa-06-slot-census/formal_h20/result/receipts/terminal-files.sha256`
Independent preproducer slot census	`evidence/experiment_registry.json`, entry `E-SLOT-CENSUS-A`; execution package, formal archive, preexecution receipt, aggregate, controls, and terminal ledger are path- and SHA-bound there
Dual-producer repeat	`evidence/experiment_registry.json`, entry `E-DUAL-PRODUCER-REPEAT-A`; execution package, formal archive/sidecar, summary digest, and exact replay boundary are bound there
Falcon-H1 bounded transfer	`evidence/experiment_registry.json`, entry `E-FALCON-H1-TRANSFER-V2-A`; formal archive/sidecar, count clarification, validation, aggregate, and terminal ledger are path- and SHA-bound there
Primary compiled dispatch	`evidence/experiment_registry.json`, entry `E-COMPILED-DISPATCH-A`; the immutable source package retains its pre-run HOLD metadata, while the separately hash-bound post-run aggregate and mirror authorize the bounded target-5 result
Selected-cell live binding	`evidence/experiment_registry.json`, entry `E-LIVE-BINDING-A`, and `package/fa-09-live-binding-audit-mirror/`; the immutable package retains pre-run metadata, while the terminal result and hash-identical post-run audit authorize only the bounded selected-cell claim

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Table 32 continued.

Claim/evidence	Machine-readable source
Designer–executor-separated faults	`package/fa-05-separated-faults/author_freeze/designer_input/prior-submission.pdf` `package/fa-05-separated-faults/author_freeze/PROTOCOL.md` `package/fa-05-separated-faults/executor_attempt_b/AMENDMENT.md` `package/fa-05-separated-faults/formal_h20/RESULT_VERIFICATION.json` `package/fa-05-separated-faults/formal_h20/r33-fresh-faults-20260825b/rank-run-*/rank-*/mutant-case.json` `package/fa-05-separated-faults/formal_h20/r33-fresh-faults-20260825b/summary.json` `package/fa-05-separated-faults/formal_h20/r33-fresh-faults-20260825b/terminal-files.sha256`
Historical alias regression and repair	`package/fa-07-alias-regression/design_decision.md` `package/fa-07-alias-regression/preregistration.json` `package/fa-07-alias-regression/preexecution-freeze-receipt.json` `package/fa-07-alias-regression/preexecution-resource-amendment.json` `package/fa-07-alias-regression/formal_h20/RESULT_VERIFICATION.json` `package/fa-07-alias-regression/formal_h20/r35-historical-alias-20260826a/aggregate.json` `package/fa-07-alias-regression/formal_h20/r35-historical-alias-20260826a/replay/`
Bounded two-stream lifecycle	`package/fa-03-two-stream/formal_run_20260825b/raw/formal-result.json` `package/fa-03-two-stream/formal_run_20260825b/replay/independent-replay.json` `package/fa-03-two-stream/formal_run_20260825b/receipts/raw-and-replay.sha256`
Expanded captured-boundary numerical sweep	`package/fa-04-numerical-sweep/`; 272 local arrays under `raw/sidecars/`; canonical result `oracle-result.json`; local replay source `r30_expanded_oracle_reference.py`; validation `validation_report.json`; terminal binding `terminal-products.sha256`
Local replay and storage accounting	`evidence/experiment_registry.json`, entry `E-LOCAL-REPLAY-STORAGE-COST-V1-A`; protocol, Attempt-B amendment, aggregate, independent audit, and terminal ledger reside under `evidence/r40_ci_cost_accounting_v1/`
Composition run (packed entry shared at $N > 1$)	Per-rank preflight gate; formal per-rank shards carrying per-target coverage and predicate rows, per-request ownership byte accounting, token traces against the published private-materialization reference, and per-fanout working-set/transient fields; torch-free aggregation with a blind contract replay that re-derives every target status from the row's own coverage and predicate and reports drift as a defect
Manuscript claim index	`evidence/claim_evidence_map.tsv`